Introduction to Statistical Analysis in Agriculture and the Natural Resources

AGNR 220 / SNR 220

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Welcome!

Introduction to Statistical Analysis in Agriculture and the Natural Resources

Welcome to **AGNR 220 / SNR 220 — Introduction to Statistical Analysis in Agriculture and the Natural Resources**! This companion book supports the Fall 2026 course taught by Dr. Alejandro Molina-Moctezuma at the University of Tennessee.

Every chapter in this book pairs directly with an in-class slideshow presentation. Think of the slideshows as the "live" portion of the lesson and this book as the place you come back to for deeper explanations, step-by-step examples, and a reference you can consult whenever you need it. **This book will be updated throughout the semester** — new chapters, examples, and figures will be added as we progress through the material.

> **i** How to use this book
>
> Each chapter corresponds to one week of class. Before each week, skim the **Learning Objectives** so you know what to focus on. After class, use the **How to Do It by Hand**, **How to Do It in Excel**, and **How to Do It in R** sections to practice what you learned. The **Practice Problems** at the end of each chapter are great preparation for assignments and exams.

No prior statistics experience is required for this course — just basic arithmetic. We will build every concept from scratch using real examples from agriculture and natural resources.

Syllabus

Please check Canvas for the most up-to-date syllabus, or see the **Syllabus** chapter in the left sidebar.

Objectives and structure

Specific Course Objectives

This is an **applied** data analysis course targeted to students working in natural resources and agriculture. This course will focus on real-world problems and real-world data relevant to natural resources, animal science, plant science, and agriculture. All the applied problems we will work on are from the real world, and can be solved using real-world data. The ultimate **objective** is for students to be able to answer and solve data problems commonly found in the natural resources by analyzing data and inferring results. Some of the outcomes for students are to:

1. identify data types and statistical methods,
2. apply statistical techniques to a variety of natural resource management questions,
3. identify and apply statistical methods commonly used to answer questions in agricultural resource analysis,
4. communicate results of statistical analysis to stakeholders (e.g., users of natural and agricultural resources),
5. apply statistical knowledge to decision making in natural resources and agriculture, and
6. analyze data sets using Excel and a basic introduction to R.

Course Structure

This course is structured in five units. Each week covers one topic area and corresponds to one chapter in this book. During each week you will:

1. **Learn:** Attend the lecture and follow along with the slideshow.
2. **Practice:** Work through in-class examples collaboratively with your classmates.
3. **Apply:** Tackle a real-world agricultural or natural resource scenario.
4. **Read:** Use this book for background, step-by-step examples, and worked problems.
5. **Submit:** Complete take-home assignments that reinforce the week's skills.

Book Units

Unit	Topic Area	Weeks
1	Getting Started with Data	1–4
2	Probability and Uncertainty	5–7
3	Inference and Hypothesis Testing	8–11
4	Comparing Groups	12–14
5	Relationships Between Variables	15

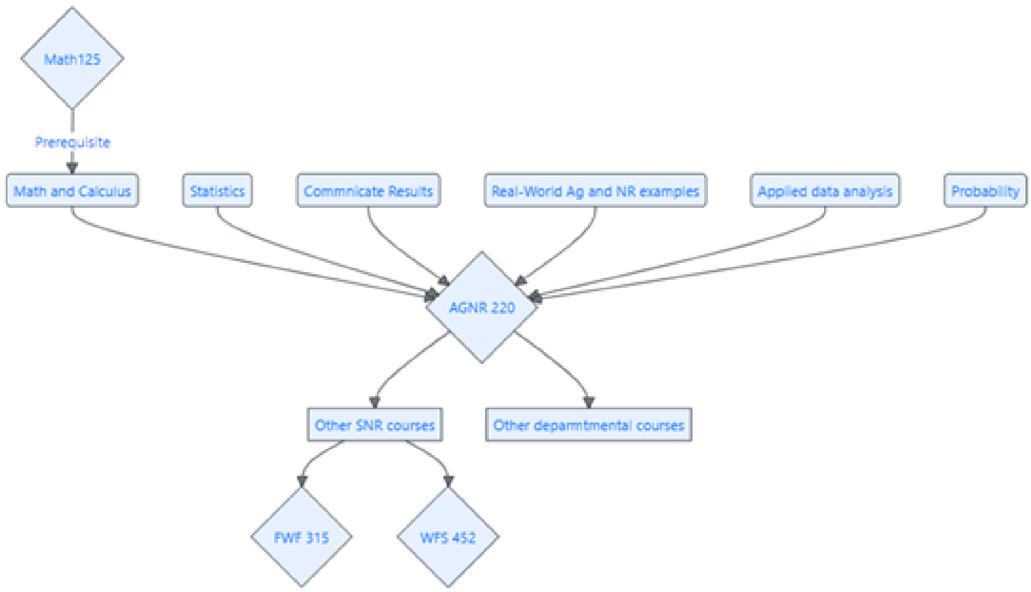


Figure 1: Course map

Citing this book

Please cite this book using:

Part I

Front Matter

1 Syllabus: AGNR 220 - Introduction to Statistical Analysis in Agriculture and the Natural Resources

2 Course Information

Item	Details
Course	AGNR 220: Introduction to Statistical Analysis in Agriculture and the Natural Resources
Cross-listed as	SNR 220
Credits	3
General Education	Quantitative Reasoning (QR)
Primary emphasis	Applied statistical analysis using agriculture and natural-resources data

2.1 Course Description

This course provides students with an overview of statistical methods used in agriculture, animal sciences, plant sciences, and natural resource management and research. Students will become familiar with data collection techniques and experimental design, probability distributions, statistical tests, measures of central tendency and dispersion, simple linear regression, correlation, and linear models.

3 Learning Outcomes

3.1 Quantitative Reasoning (QR)

Quantitative and statistical evidence and mathematical and logical reasoning often play critical roles in building arguments and claims that support opinions and actions. Students should therefore possess the mathematical and quantitative skills needed to evaluate such arguments and claims, recognize the quantitative dimensions of questions they will encounter in their professional and personal lives, and use mathematical and logical reasoning to formulate and solve problems.

Students completing this course will demonstrate the ability to:

1. Identify aspects of arguments and claims that rely on quantitative evidence and mathematical or logical reasoning.
2. Evaluate the appropriateness of conclusions drawn from quantitative evidence and mathematical or logical reasoning techniques.
3. Formulate and solve problems that rely on mathematical or logical reasoning.

These outcomes are addressed in addition to the course-specific learning outcomes below.

3.2 Course-Specific Learning Outcomes

AGNR 220 is an applied data-analysis course for students working in natural resources and agriculture. The course emphasizes real-world problems and data relevant to natural resources, animal science, plant science, and agriculture.

By the end of the course, students should be able to:

1. Identify common data types and appropriate statistical methods.
2. Apply statistical techniques to natural-resource and agricultural questions.
3. Select statistical methods appropriate for agricultural and natural-resource analyses.
4. Communicate the results of statistical analyses to stakeholders.
5. Apply statistical knowledge to decision-making in natural resources and agriculture.
6. Analyze data sets using Excel and an introductory level of R.

4 Course Structure

The course is organized into units with specific learning objectives. Most units will combine several types of activities:

Component	What to Expect
Learning	Introduction to the statistical concepts and reasoning needed for the unit.
Practice	Guided examples completed in class, often collaboratively.
Real-world application	An agriculture or natural-resources scenario in which students identify an appropriate analysis, analyze data, draw conclusions, and communicate recommendations.
Readings and discussions	Online material that supplements class activities and provides applied context.
Take-home work	Assignments based on the concepts and real-world applications developed during the unit.

5 Course Materials

5.1 Primary Text

The Primary text will be published online and is available for free. It is a companion to the course and will be updated throughout the semester. The text provides background, step-by-step examples, and worked problems. The link will be available on the course Canvas page after the first class.

5.2 Additional Recommended References

The following books are **recommended but not required**. They provide more advanced material for students with specific interests.

- Whitlock, M. C., and D. Schluter. 2020. *The Analysis of Biological Data*. Macmillan Learning. 810 pp.
- Bivand, R. S., E. Pebesma, and V. Gómez-Rubio. 2013. *Applied Spatial Data Analysis with R*. 2nd ed. Springer. 405 pp.
- Guy, C. S., and M. L. Brown. 2007. *Analysis and Interpretation of Fisheries Data*. American Fisheries Society. 961 pp.
- Lawson, J. 2014. *Design and Analysis of Experiments with R*. Chapman & Hall/CRC Texts in Statistical Science. 620 pp.
- Robinson, A. P., and J. D. Hamann. 2011. *Forest Analytics with R: An Introduction*. Springer. 339 pp.
- Stevens, M. H. H. 2009. *A Primer of Ecology with R*. Springer. 388 pp.
- Wickham, H., and G. Grolemund. 2017. *R for Data Science*. O'Reilly Media. 522 pp.

Additional readings will include articles on data analysis and applied studies in agriculture and natural resources. These readings will either be open access or available through the University library.

5.3 Course Concept Map

The course concept map represents the major knowledge areas that inform AGNR 220 and the courses that build on knowledge developed here. The concept map may be updated as the course develops.

6 Grading

6.1 Point Distribution

Assessment	Points	Percent of Final Grade
Assignments	400	40%
Take-home exam	100	10%
Midterms (2)	200	20%
Critical-thinking assignments (3)	150	15%
Final exam	150	15%
Total	**1000**	**100%**

Approximately 25 points of extra credit may be available during the semester.

6.2 Grading Scale

Letter Grade	Points	Percentage
A	>914	>91.49%
A-	900–914	90–91%
B+	880–899	88–89%
B	820–879	82–87%
B-	800–819	80–81%
C+	780–799	78–79%
C	720–779	72–77%
C-	700–719	70–71%
D	600–699	60–69%
F	<600	<60%

7 Course Units

The course is organized into five broad units. The week ranges below are **guides only** and are intended to show the general progression of the course. The actual timing of each unit may change depending on how students are mastering the material. We may spend more time on some topics when additional practice is needed or move more quickly when the class is ready to progress.

Unit	Topic Area	Approximate Weeks
1	Getting Started with Data	1–3
2	Probability and Uncertainty	4–6
3	Inference and Hypothesis Testing	7–10
4	Comparing Groups	11–13
5	Relationships Between Variables	14–15

8 Assignments and Assessments

8.1 Weekly Assignments and Exams

The core of the questions in weekly assignments and exams is the connection between a real problem or question and a potential solution. Many problems will require multiple quantitative and logical-reasoning steps rather than the use of a single formula or procedure.

8.2 Take-Home Exam

The take-home exam requires students to investigate a problem and use the quantitative and analytical tools developed in the course to address a real-world natural-resources or agricultural management question.

8.3 Critical-Thinking Assignments

Critical-thinking assignments ask students to connect real-world problems, quantitative methods, and logical reasoning to make recommendations about issues in natural resources or agriculture.

These assignments may have multiple defensible solutions. Students are expected to think critically about the evidence supporting their recommendation and the implications of the decisions they make.

9 Due Dates and Late Work

9.1 Due Dates

Weekly assignments will generally be due seven days after they are published. Specific due dates will be listed in Canvas.

9.2 Late Policy

Students are expected to submit assignments and papers on time. However, life situations can occasionally make delays unavoidable.

Each student is allowed **one seven-day extension**, no questions asked and with no points deducted.

After that extension has been used, the following deductions apply:

How Late?	Deduction
Up to 7 days late	-10%
8–14 days late	-25%
More than 14 days late	Assignment will not be accepted

All late work must be submitted before the beginning of final-exam week.

9.2.1 Exceptions

Extensions for illness or other justified personal circumstances will be considered on a case-by-case basis. Please contact the instructor as soon as possible when circumstances affect your ability to complete course work.

10 Communication

Canvas and UT email will be the primary methods of course communication. Students are expected to check their UT email frequently, ideally at least every other day, to keep up with class assignments, announcements, and activities.

11 Academic Integrity and Use of Artificial Intelligence

Academic dishonesty of any sort will not be tolerated. Plagiarism includes copying answers, phrases, portions of sentences, or main ideas from another person, including classmates or students who previously completed the course, on work submitted for a grade.

In this course, submitted work is expected to be produced by the students themselves, whether completed individually or collaboratively as directed. Students must not use generative AI tools such as ChatGPT to generate answers or text for submitted course work. Use of a generative AI tool to complete an assignment constitutes academic dishonesty.

12 Changes to the Syllabus

We plan to follow the syllabus as laid out here. If changes are needed, students will be notified of updates to the course calendar, assignments, grading policies, or other course requirements.

13 University Policies

13.1 Academic Honesty and Student Conduct

When you enroll at the University of Tennessee, you pledge to hold yourself and your peers to the standards of academic honesty and integrity described in the University's Academic Policies and Code of Conduct. Academic dishonesty in any form will not be tolerated.

For additional information, see the University of Tennessee Student Code of Conduct.

13.2 Student Disability Services

The University of Tennessee, Knoxville, is committed to providing an inclusive learning environment for all students. If you anticipate or experience a barrier in this course due to a chronic health condition, learning, hearing, neurological, mental-health, vision, physical, or other disability, or because of a temporary injury, you are encouraged to contact Student Disability Services (SDS) at 865-974-6087 or sds@utk.edu.

An SDS Coordinator will meet with you to develop a plan to ensure equitable access to this course. If you are already registered with SDS, please contact the instructor to discuss implementation of the accommodations included in your course access letter.

Part II

Unit 1: Getting Started with Data

14 Week 1 - Welcome and What Is Data?

Dates: Monday August 17 · Wednesday August 19 · Friday August 21

14.1 Learning Objectives

By the end of this week, you should be able to:

- Explain why data and statistics are important in agriculture and natural resources.
- Define the terms *variable*, *observation*, and *dataset*.
- Classify variables as categorical or numerical (and within each type, name the subtypes).
- Give at least two real-world agricultural or natural resource examples for each data type.
- Describe how this course is organized and what to expect each week.

14.2 Why Do We Use Data and Statistics?

Every day, people working in agriculture and natural resources face questions that data can help answer:

- Is this new corn variety producing higher yields than the old one?
- Are deer populations in this county growing, stable, or declining?
- Does adding fertilizer to these fields actually improve grass production, or does the variation between fields just make it look that way?
- Is the water in this stream safe for fish?

Without data, we can only guess at the answers. With data - and the right tools to analyze them - we can make confident, defensible decisions.

> **Think About It**
>
> Think of a decision that a farmer, a wildlife manager, or a water quality specialist might make in a typical week. What information would they need to make a *good* decision rather than just an educated guess? Where would that information come from?

Statistics is the science of collecting, organizing, analyzing, and interpreting data. In this course, you will learn how to do all four of those steps - first by hand (so you truly understand what is happening), then in Excel, and then in R.

14.2.1 Real-World Examples of Statistics in Action

Food Security and Crop Production Plant scientists use statistics to test whether new crop varieties yield more grain per hectare than current varieties. A single farmer's field is not enough evidence - we need data from many fields, many locations, and multiple growing seasons. Statistical analysis tells us whether an observed difference is real or could have happened by random chance.

Wildlife Management Wildlife managers count deer, elk, or turkey populations to set hunting regulations. A simple count from one area on one day is misleading - animals move, observers miss individuals, and conditions vary. Statistics allows managers to estimate population size with a known margin of error.

Water Quality Environmental scientists monitor streams for pollutants such as nitrates from fertilizer runoff. They collect water samples at multiple times and locations. Statistics helps them determine whether nitrate levels exceed safe thresholds and whether levels are changing over time.

Livestock Performance Animal scientists compare weight gain in steers on different diets. Is the difference in average weight gain between two groups due to the diet, or just random variation among individual animals? A statistical test answers that question.

14.3 What Is a Dataset?

When we collect information, we organize it into a **dataset**. A dataset is a structured collection of data about a group of *subjects* (also called *units* or *cases*).

Consider this small example:

Steer ID	Diet	Daily Gain (kg)	Age (months)	Breed
1	Standard	1.2	18	Angus
2	Supplement	1.5	20	Hereford
3	Standard	1.1	19	Angus
4	Supplement	1.6	21	Angus
5	Standard	1.3	17	Hereford

Each **row** is one *observation* - in this case, one steer. Each **column** is one *variable* - a characteristic measured on each steer.

Key Vocabulary

Dataset: A table of data with rows (observations) and columns (variables).
Observation: One subject/unit in your study (one steer, one field, one water sample, one tree).
Variable: A characteristic measured or recorded for each observation (weight, species, temperature, yield).

14.4 Types of Variables

Not all variables are the same kind. The type of variable determines what statistics you can use and what kind of plots make sense. There are two main families:

14.4.1 Categorical Variables

A **categorical variable** places each observation into one of a fixed set of categories. You cannot do arithmetic on categories (what is the "average" of Angus + Hereford?).

Nominal variables - categories with no natural order. Examples: - Crop species: corn, soybeans, wheat, sorghum - Land cover type: forest, pasture, cropland, wetland - Sex: male, female - Treatment group: control, low fertilizer, high fertilizer

Ordinal variables - categories that do have a natural order, but the gaps between levels are not equal. Examples: - Soil compaction rating: low, medium, high - Body condition score of a cow: 1, 2, 3, 4, 5 (numbers used as ranks) - Wildlife habitat quality: poor, fair, good, excellent

Think About It

If a researcher rates a stream's water quality as "1 = poor, 2 = fair, 3 = good," is the difference between "poor" and "fair" the same as the difference between "fair" and "good"? Why does that matter for choosing a statistical method?

14.4.2 Numerical Variables

A **numerical variable** takes actual number values, and arithmetic on those numbers is meaningful (a steer that weighs 400 kg really does weigh twice as much as one that weighs 200 kg).

Discrete variables - whole-number counts. You cannot have 2.5 deer. Examples: - Number of eggs in a bird nest - Count of seedlings that germinated in a plot - Number of insects caught in a trap

Continuous variables - can take any value in a range, including decimals. Examples: - Tree diameter at breast height (cm) - Soil pH (e.g., 6.3, 7.1, 5.8) - Precipitation in millimeters - Milk production per cow per day (liters) - Temperature in degrees Celsius

> **Summary of Variable Types**

```
Variables
  Categorical
     Nominal (no order): crop species, land use type
     Ordinal (ordered): habitat quality, body condition score
  Numerical
     Discrete (counts): number of deer, seedling germination count
     Continuous (measurements): tree diameter, milk yield, soil pH
           Discrete (counts): number of deer, seedling germination count
           Continuous (measurements): tree diameter, milk yield, soil pH
```

14.5 Overview of the Course Structure

This course is organized into five units. Each week covers one main topic. Below is the roadmap for the semester.

Unit	Weeks	Topics
1 - Getting Started with Data	1–4	Data types, describing data, uncertainty
2 - Probability and Uncertainty	5–7	Probability, frequency distributions, binomial
3 - Inference and Hypothesis Testing	8–11	NHST framework, t-tests
4 - Comparing Groups	12–14	ANOVA, chi-square

Unit	Weeks	Topics
5 - Relationships	15	Correlation and regression
Wrap-Up	16	Review and synthesis

What to expect each week:

Each chapter in this book follows the same structure:

1. **Learning Objectives** - what you should know by Friday
2. **Background** - plain-language explanations with ag/NR examples
3. **Key Concepts** - definitions you should memorize
4. **How to Do It by Hand** - step-by-step worked examples
5. **How to Do It in Excel** - practical spreadsheet instructions
6. **How to Do It in R** - beginner-friendly code with comments
7. **Practice Problems** - problems to try on your own

14.6 Introduction to the Tools

In this course you will use two software tools:

Microsoft Excel - You almost certainly have Excel already. We use it for calculations, simple statistics, and creating basic plots. Excel is the tool of choice for many agricultural professionals and is excellent for organizing small to medium-sized datasets.

Program R - R is a free, open-source programming language designed specifically for statistical analysis. It is widely used in research, government agencies, and industry. Do not be intimidated - we will start with very simple commands and build from there.

> **!** Software Installation
>
> You will need both Excel and R/RStudio installed by **Week 2**. Instructions for installing R and RStudio are available from the instructor on Canvas. R can be downloaded for free from https://cran.r-project.org/ and RStudio from https://posit.co/download/rstudio-desktop/. If you have a Chromebook or iPad, a free cloud version of RStudio is available at https://posit.cloud.

We will use Excel first (Weeks 2–3) and introduce R gradually starting in Week 3. By the end of the semester you will be comfortable with basic data analysis in both tools.

14.7 Summary

- Statistics helps us make sound decisions in agriculture and natural resources by turning raw data into reliable conclusions.
- A dataset is a table of observations (rows) and variables (columns).
- Variables are either **categorical** (nominal or ordinal) or **numerical** (discrete or continuous).
- Knowing the type of a variable is the first step in choosing the right analysis.
- This course progresses from describing data → understanding probability → testing hypotheses → comparing groups → modeling relationships.

15 Assignment 1

Question 1

Read the four steps to take a random sample, sample of convenience and volunteer bias. Then come up with one example of each that is relevant to your major or career interest. Be prepared to share your examples in class.

15.1 Question 2

Answer the practice problems at the end of the chapter. You may work with a partner, but each student must submit their own answers. No use of AI tools is allowed for this assignment. You may use your notes, the textbook, and other course materials. You get full credit for honest attempts, even if your answers are not correct.

15.2 1. Four steps to take a random sample

A **random sample** gives every individual in the population a known and equal chance of being selected.

1. **Define the population.** Clearly identify the group you want to study. For example, all ponds within a wildlife management area.
2. **Create a sampling frame.** Make a list of all individuals or sampling units in the population, such as a list of every pond in the management area.
3. **Randomly select the sample.** Use a random number generator or another random method to select ponds from the complete list.
4. **Sample the selected units.** Collect data from the ponds that were randomly selected. Avoid replacing difficult-to-sample ponds with easier ones, since this could introduce bias.

15.3 2. Sample of convenience

A **sample of convenience** includes individuals or locations because they are easy to access rather than because they were randomly selected.

For example, a researcher studying fish abundance in a river might sample only locations near road crossings because they are easy to reach. These locations may not represent the entire river, so conclusions based on them could be biased.

15.4 3. Volunteer bias

Volunteer bias occurs when individuals choose whether to participate in a study and those who volunteer differ from those who do not.

For example, a survey asking landowners about attitudes toward prescribed fire might receive more responses from landowners who have strong opinions about prescribed fire. Their responses may therefore not accurately represent the opinions of all landowners.

15.5 Practice Problems

1. A forest ecologist records the following information for each tree in a study plot: species, diameter at breast height (cm), canopy condition (healthy, stressed, dying), and the number of woodpecker cavities present. Classify each of these four variables as nominal, ordinal, discrete, or continuous.
2. A poultry farmer wants to know whether a new feed supplement increases egg production. She records, for each hen: feed type (standard or supplement), hen age in weeks, number of eggs laid per week, and flock ID. Identify the observations and variables. Classify each variable.
3. Give one example of a dataset you might collect in your own major or career interest. List at least three variables and classify each one.

16 Week 2, Part 1: Describing Data in R

16.1 What you will do

In class, you will:

1. Create an RStudio Project.
2. Create a Quarto document that renders to a Word document.
3. Practice a few basic R commands.
4. Import the systolic blood pressure dataset used in our previous class.
5. Use R to calculate the mean, median, first quartile, third quartile, standard deviation, variance, and coefficient of variation.
6. Create a histogram.
7. Submit

After class, you will:

1. Read the rest of this chapter.
2. Complete only the sections labeled **Assignment question**.
3. Write your R code and answers in your Quarto document.
4. Render the document to Word and submit the `.docx` file.

> **!** What should you submit?
>
> Today, you should submit one rendered Word document that includes your name, and answers.

16.2 Part 1: Set up your work

16.2.1 Create an RStudio Project

Projects keep your data, code, and output together. They also make it easier for R to find your files.

! Where to put the project

You will be downloading ALL DATA FILES to this folder. Please make it an easy to find folder

1. Open RStudio.
2. Select **File > New Project > New Directory > New Project**.
3. Name the project `SNR220` –> You will use this for the rest of the semester
4. Choose a location on your computer that you can find easily.
5. Create the project.
6. Save the systolic blood pressure `.csv` file inside the project folder.

16.2.2 Create a Quarto document

1. Select **File > New File > Quarto Document**.
2. Enter a title such as `Week 2: Describing Data`.
3. Enter your name as the author.
4. Select **Word** as the output format.
5. Select **Create**.
6. Save the file as `week2_describing_data.qmd` inside your project folder.

The top of your file should look similar to this:

```
---
title: "Week 2: Describing Data"
author: "Your name"
format: docx
---
```

16.2.3 Add code and written answers

You will complete all work in the `.qmd` file. Add regular text for your written answers. Add executable R code chunks for your code.

To insert a code chunk, select **Insert > Executable Cell > R**. You can also type three backticks followed by `{r}`, add your code, and close the chunk with three more backticks.

For example:

````
### My first calculation

The result means that...


::: {.cell}

```{.r .cell-code}
2 + 2
```

::: {.cell-output .cell-output-stdout}

```
[1] 4
```



:::
:::
````

To run one line or a selected section of code, press **Ctrl + Enter** on Windows or **Command + Enter** on macOS. To create the Word document, click **Render**.

16.3 Part 2: Play with R

R can be used as a calculator. In your Quarto document, create an R code chunk and run each of the following commands one at a time.

```r
5 + 5

8^2

sqrt(9)

10 / 4
```

R can also store values in objects. The assignment operator `<-` places a value inside an object.

```r
x <- 17

y <- x * 2

x

y
```

A vector is a collection of values. The function `c()` combines values into a vector.

```r
practice_values <- c(5, 7, 9, 15, 50)

practice_values

mean(practice_values)

sd(practice_values)
```

Change one of the values and run the code again. What happens to the mean and standard deviation?

> **A note about the Console**
>
> The Console is useful for quick experiments, but your assignment code should be written in the Quarto file. Code saved in your Quarto file can be checked, corrected, rerun, and included in your final Word document.

16.4 Part 3: Guided activity with the systolic blood pressure data

You previously estimated descriptive statistics for the systolic blood pressure dataset. Now you will use R to calculate the same values.

16.4.1 Import the data

Download the dataforR.csv file from Canvas

Create an R code chunk in your Quarto document and enter:

```r
blood <- read.csv("systolicBloodPressure.csv")
```

The assignment operator `<-` stores the imported dataset in an object named `blood`.

Use these functions to inspect the dataset:

```r
head(blood)

names(blood)

str(blood)

summary(blood)
```

- `head()` displays the first few observations.
- `names()` displays the variable names.
- `str()` shows the structure and variable types.
- `summary()` provides a quick summary of each variable.

16.4.2 Calculate the descriptive statistics

Use the following functions to estimate the function in your Quarto document:

```r
mean(blood)

median(blood)

quantile(blood, probs = 0.25)

quantile(blood, probs = 0.75)

sd(blood)

var(blood)
```

$$CV = \frac{s}{\bar{y}} \times 100$$

Calculate it in R:

```r
sd(blood / mean(blood) * 100
```

> **! Before you render and submit**
>
> Make sure your values are the same as you got on Excel

Now, answer in the Quarto document:

Was it easy or hard to do this?

16.4.3 Create a quick histogram

A histogram displays the distribution of a numerical variable. It can help you see the center, spread, skewness, gaps, and possible extreme values.

Add this code to your Quarto document:

```r
hist(
  blood,
  main = "Distribution of systolic blood pressure",
  xlab = "Systolic blood pressure (mm Hg)",
  col = "lightblue",
  border = "white"
)
```

The histogram will appear automatically in the rendered Word document. Below the plot, write two or three sentences describing what you see. Consider the center, spread, shape, and any possible extreme values.

> **⚠ Warning**
>
> That is it! You get to go home. The assignment will be due Wednesday 2nd! After we are back from labor day!

17 Part 4: Concepts to review

17.0.1 Center and spread

Raw observations can be difficult to interpret. Descriptive statistics reduce a dataset to a few useful summaries.

We usually want to know:

1. Where is the center of the distribution?
2. How spread out are the observations?

Both are important. Two groups can have the same mean but very different amounts of variation.

17.0.2 Measures of center

Mean: Add all observations and divide by the number of observations. The mean uses every value, so extreme values can strongly affect it.

Median: Sort the observations and find the middle value. The median is less affected by extreme values and is often useful for skewed data.

Mode: The value that occurs most often. It is especially useful for categorical variables and discrete counts. A dataset can have more than one mode or no unique mode.

> Think about it
>
> A soil scientist measures nitrate concentration in eight stream samples. One sample has a much higher concentration than the others. How might that observation affect the mean? How might it affect the median? Which value would better describe a typical sample?

17.0.3 Quartiles

After sorting the data:

- The first quartile, Q1, is the 25th percentile.
- The median, Q2, is the 50th percentile.
- The third quartile, Q3, is the 75th percentile.

The middle 50 percent of the observations fall between Q1 and Q3.

17.0.4 Variance and standard deviation

A deviation is the distance between an observation and the mean:

$$\text{deviation}_i = y_i - \bar{y}$$

Some deviations are positive and some are negative. If we simply add them, they cancel and sum to zero. Variance avoids this problem by squaring the deviations.

For a sample, the variance is:

$$s^2 = \frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n-1}$$

The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n-1}}$$

Variance is expressed in squared units. Standard deviation is expressed in the original units, so it is usually easier to interpret. A larger standard deviation indicates greater variation around the mean.

> **!** Why use n - 1?
>
> When we use a sample to estimate variation in a larger population, sample variance divides by $n-1$. This correction improves the estimate of population variance. For now, remember that `var()` and `sd()` in R use $n-1$ automatically.

17.0.5 Coefficient of variation

The coefficient of variation expresses the standard deviation relative to the mean. Because it is a percentage, it can help compare variation among measurements with different means.

For example, a CV of 12 percent means that the standard deviation is 12 percent of the mean.

Use the CV only when a value of zero is meaningful and the mean is positive. It is not appropriate for every type of measurement.

17.0.6 A short example

A rancher records daily weight gain for 10 steers:

```
1.2, 1.5, 1.3, 1.8, 1.1, 1.4, 1.6, 1.2, 1.7, 1.4
```

The main summaries are:

Statistic	Result
Mean	1.42 kg/day
Median	1.40 kg/day
Q1	1.225 kg/day
Q3	1.575 kg/day
Variance	0.0529 $(\text{kg/day})^2$
Standard deviation	0.230 kg/day
CV	16.2%

The mean and median are similar, which suggests that the center is not being strongly affected by an extreme value. The standard deviation describes the typical spread around the mean. It does not guarantee that every observation will fall within one standard deviation of the mean.

17.1 Excel reference only

> **ℹ You do not have to complete this section**
>
> This section is included as a reference. Do not submit Excel calculations for this assignment. Complete the assignment questions using R.

If values are stored in cells `A1:A10`, Excel can calculate the following:

Statistic	Excel formula
Mean	`=AVERAGE(A1:A10)`
Median	`=MEDIAN(A1:A10)`
First quartile	`=QUARTILE.INC(A1:A10,1)`
Third quartile	`=QUARTILE.INC(A1:A10,3)`
Sample variance	`=VAR.S(A1:A10)`
Sample standard deviation	`=STDEV.S(A1:A10)`
Coefficient of variation	`=STDEV.S(A1:A10)/AVERAGE(A1:A10)*100`

Use `VAR.S` and `STDEV.S` when the observations are a sample. The functions `VAR.P` and `STDEV.P` are used when the data include the entire population of interest.

17.2 Homework: R exercises

Complete both assignment questions in your Quarto document. Include the code, output, plots, and written responses. Do not calculate these answers in Excel.

17.2.1 Assignment question 1: Deer observations

A wildlife biologist records the number of white-tailed deer observed during eight transect surveys:

```
4, 7, 3, 9, 5, 6, 4, 10
```

1. Create a vector named `deer` containing these observations.
2. Write R code to calculate the mean, median, Q1, Q3, range, variance, standard deviation, and CV.
3. Create a histogram with an informative title and axis label.
4. In two or three sentences, describe the center and spread of the observations. Mention whether the mean and median are similar.

> Remember
>
> The R function `range()` returns the minimum and maximum. To calculate the difference between them, use `diff(range(deer))`.

17.2.2 Assignment question 2: Wheat yield and an extreme value

A farmer records wheat yield, in bushels per acre, for six fields:

```
52, 48, 55, 61, 47, 50
```

A seventh irrigated field yields 110 bushels per acre.

1. Create a vector named `wheat` containing the original six observations.
2. Create a second vector named `wheat_with_extreme` that includes all seven observations.
3. For each vector, write R code to calculate the mean, median, and standard deviation.
4. Create one histogram for each vector. Give each histogram a clear title and use the same x-axis limits so the plots can be compared fairly. You can add `xlim = c(40, 115)` inside each `hist()` function.
5. In three or four sentences, explain how the value of 110 affected the mean, median, standard deviation, and shape of the distribution.

17.3 Before submitting

Check that your Word document includes:

- Your name and assignment title
- All R code used to answer the assignment questions
- The numerical output from R
- All required histograms
- Written interpretations in complete sentences

Click **Render** and open the `.docx` file to make sure the code, output, and plots are visible before submitting it.

18 Week 3 - Describing Data, Part II: Distributions, Histograms, and Boxplots

Dates: Monday August 31 · Wednesday September 2 · Friday September 4

18.1 Learning Objectives

By the end of this week, you should be able to:

- Draw and interpret a histogram from a small dataset.
- Describe the shape of a distribution (symmetric, right-skewed, left-skewed).
- Calculate quartiles and the interquartile range (IQR).
- Draw and interpret a boxplot by hand.
- Create histograms and boxplots in Excel and R.

18.2 Background and Conceptual Introduction

Last week we calculated summary numbers - the mean, median, and standard deviation. Those numbers are useful, but they do not tell the whole story. Two datasets can have the same mean and standard deviation yet look completely different when plotted.

Think of it this way: knowing that a forest has an average tree diameter of 20 cm is useful. But knowing that the diameters are spread roughly evenly around 20 cm (bell-shaped) is very different from knowing that most trees are small saplings and a few are giant old-growth trees (right-skewed). The *shape* of the distribution matters for choosing the right statistical test.

Histograms and **boxplots** are two of the most important tools for visualizing the distribution of a numerical variable. They let you see the shape, identify typical values, spot unusual observations (outliers), and compare groups.

Think About It

You are comparing fish length data from two different lakes. Lake A has an average trout length of 35 cm with most fish close to that size. Lake B also has an average of 35 cm, but it has a mix of very small young fish and a few very large old fish. If you only reported the mean, would someone reading your report notice the difference? What would a histogram show that the mean cannot?

18.3 Key Concepts and Definitions

Frequency: The number of observations that fall within a given interval or category.

Histogram: A bar chart for numerical data. Each bar covers a range of values (a "bin"), and the height of the bar shows how many observations fall in that range. Bars touch each other because the data are continuous.

Bin (class interval): The range of values covered by one bar in a histogram. Choosing the number of bins affects how a histogram looks.

Distribution: The overall pattern of values in a dataset - where values cluster, how spread out they are, and whether they are symmetric or skewed.

Symmetric distribution: The left and right halves of the histogram mirror each other. The mean and median are approximately equal.

Right-skewed (positively skewed): The histogram has a long tail extending to the right. Most values are small, but a few are very large. The mean is pulled to the right of the median. Common in income data, species abundance counts, and pollutant concentrations.

Left-skewed (negatively skewed): The histogram has a long tail extending to the left. The mean is pulled to the left of the median.

Quartiles: Values that split a sorted dataset into four equal parts. - Q1 (first quartile) = 25th percentile - Q2 (second quartile) = median = 50th percentile - Q3 (third quartile) = 75th percentile

Interquartile Range (IQR): $IQR = Q3 - Q1$. It describes the spread of the middle 50% of the data. It is resistant to outliers.

Outlier: An observation that is far from the rest. A common rule: any value more than $1.5 \times IQR$ below Q1 or above Q3 is considered a potential outlier.

Boxplot (box-and-whisker plot): A graphical display showing Q1, median, Q3, and the range of non-outlier values. Outliers are plotted as individual points.

18.4 How to Do It by Hand

> **Example: Soil Nitrate Concentrations**
>
> A water quality specialist measures soil nitrate concentration (mg/kg) at 12 sampling locations across a farm field:
> **Data:** 8, 14, 11, 22, 9, 16, 13, 19, 10, 15, 12, 17

18.4.1 Building a Histogram

Step 1: Sort the data

8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 19, 22

Step 2: Choose bins

Range $= 22 - 8 = 14$. With 12 observations, 3–4 bins works well. Use bins of width 4:

Bin	Range	Observations	Frequency
1	8–11	8, 9, 10, 11	4
2	12–15	12, 13, 14, 15	4
3	16–19	16, 17, 19	3
4	20–23	22	1

Step 3: Draw the histogram

Draw a horizontal axis (Nitrate concentration), a vertical axis (Frequency), and draw a bar for each bin with height equal to its frequency.

18.4.2 Building a Boxplot

Step 1: Find Q1, median (Q2), Q3

Sorted: 8, 9, 10, 11, **12, 13** | **14, 15**, 16, 17, 19, 22

Median (Q2) = average of 6th and 7th values $= \frac{13+14}{2} = 13.5$

Lower half: 8, 9, 10, 11, 12, 13 $\rightarrow$ Q1 = average of 3rd and 4th $= \frac{10+11}{2} = 10.5$

Upper half: 14, 15, 16, 17, 19, 22 $\rightarrow$ Q3 = average of 3rd and 4th $= \frac{16+17}{2} = 16.5$

Step 2: Calculate IQR

$$IQR = Q3 - Q1 = 16.5 - 10.5 = 6.0$$

Step 3: Calculate fence values (outlier boundaries)

Lower fence $= Q1 - 1.5 \times IQR = 10.5 - 9.0 = 1.5$

Upper fence $= Q3 + 1.5 \times IQR = 16.5 + 9.0 = 25.5$

All observations fall between 1.5 and 25.5, so there are **no outliers**.

Step 4: Draw the boxplot

- Draw a box from Q1 (10.5) to Q3 (16.5)
- Draw a line inside the box at the median (13.5)
- Draw whiskers from Q1 down to the minimum (8) and from Q3 up to the maximum (22)

18.5 How to Do It in Excel

18.5.1 Histogram

1. Enter your data in column A.
2. Click **Insert** → **Charts** → select the **Histogram** icon (it looks like vertical bars).
3. Excel will automatically create a histogram. Right-click on the bars and select **Format Data Series** to adjust the number of bins.
4. Label your axes: right-click each axis, choose **Add Axis Title**.

> **i** Note
>
> Screenshot will be added here.

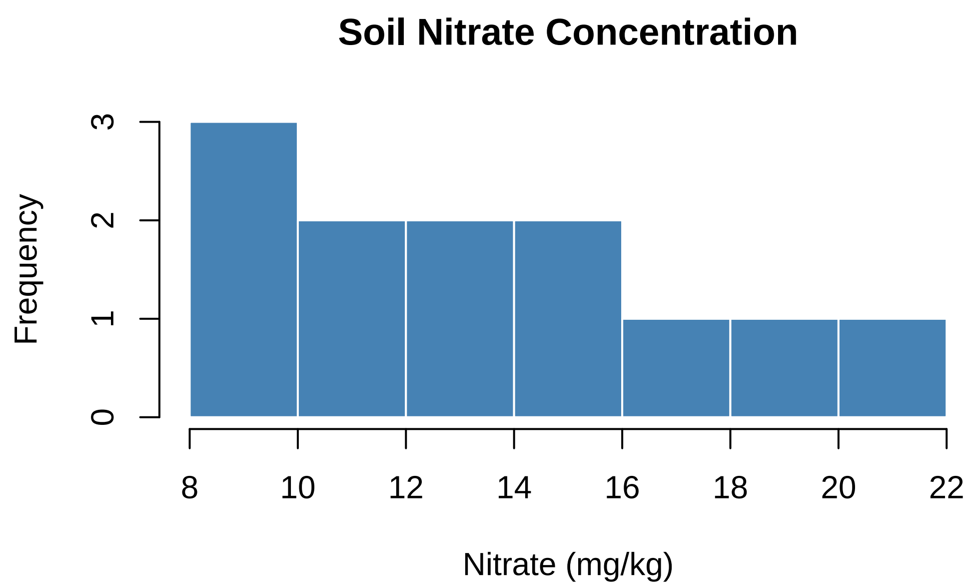
18.5.2 Boxplot

1. Select your data column.
2. Click **Insert** → **Charts** → **Insert Statistic Chart** → **Box and Whisker**.
3. Excel draws the boxplot. Note that Excel calculates quartiles slightly differently from how we did it by hand - small differences are normal.

> **Note**
>
> Screenshot will be added here.

18.6 How to Do It in R

```
# Enter the nitrate data
nitrate <- c(8, 14, 11, 22, 9, 16, 13, 19, 10, 15, 12, 17)

# --- Histogram ---
hist(nitrate,
     main = "Soil Nitrate Concentration",
     xlab = "Nitrate (mg/kg)",
     ylab = "Frequency",
     col  = "steelblue",
     border = "white")
```



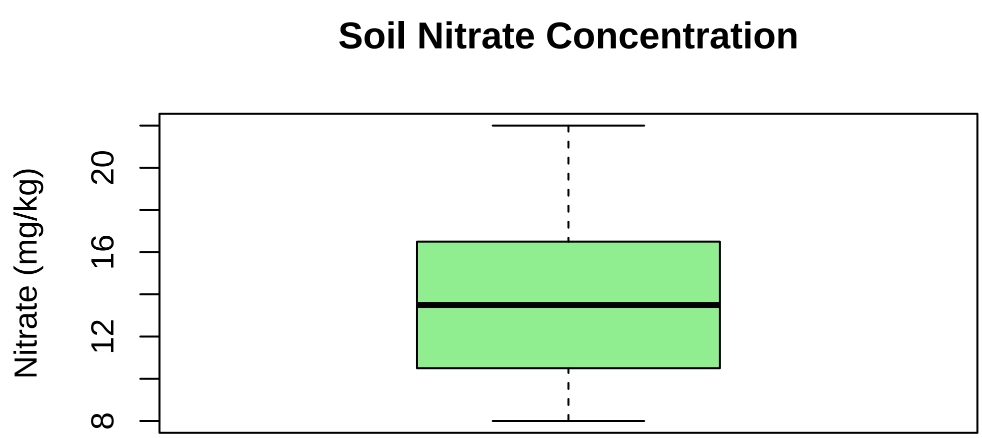
```r
# --- Quartiles and IQR ---
quantile(nitrate)          # shows min, Q1, median, Q3, max
```

```
   0%   25%   50%   75%  100% 
 8.00 10.75 13.50 16.25 22.00 
```

```r
IQR(nitrate)               # interquartile range
```

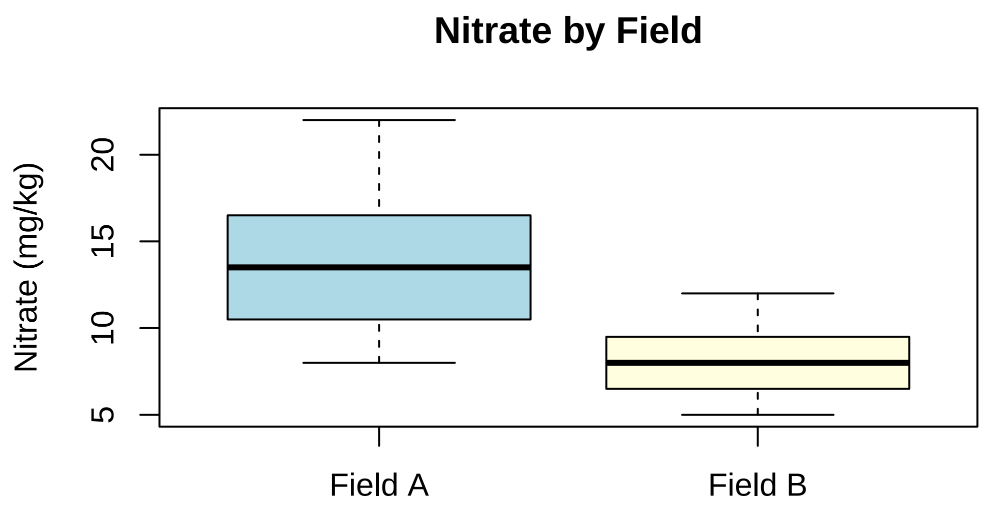
```
[1] 5.5
```

```r
# --- Boxplot ---
boxplot(nitrate,
        main   = "Soil Nitrate Concentration",
        ylab   = "Nitrate (mg/kg)",
        col    = "lightgreen")
```




```r
# --- Side-by-side boxplots for two groups (example) ---
# Suppose we have data from Field A and Field B
field_A <- c(8, 14, 11, 22, 9, 16, 13, 19, 10, 15, 12, 17)
field_B <- c(5, 7, 6, 9, 8, 11, 6, 10, 7, 8, 9, 12)
```

```r
boxplot(field_A, field_B,
        names  = c("Field A", "Field B"),
        main   = "Nitrate by Field",
        ylab   = "Nitrate (mg/kg)",
        col    = c("lightblue", "lightyellow"))
```



18.7 Interpretation and Common Mistakes

Reading a boxplot:

- A box that is taller (larger IQR) = more variable middle 50%
- If the median line is not in the center of the box, the data is skewed
- Long upper whisker or outlier points on top = right-skewed
- Long lower whisker or outlier points on bottom = left-skewed

Comparing groups:

When you put two boxplots side by side, overlapping boxes suggest the groups are similar; non-overlapping boxes suggest a meaningful difference. (We will test this formally with statistical tests in Units 3 and 4.)

Common Mistake: Connecting Histogram Bars

In a bar chart for categorical data, bars are separated by spaces. In a histogram for continuous numerical data, bars should touch. If your histogram has gaps between bars, check that you have the right chart type.

Common Mistake: Too Few or Too Many Bins

With too few bins you lose detail; with too many bins the histogram becomes jagged and noisy. A good rule of thumb: use $\sqrt{n}$ bins where n is the number of observations. For 12 observations, $\sqrt{12} \approx 3.5$, so 3–4 bins is sensible.

Common Mistake: Confusing Standard Deviation with IQR

Both measure spread, but in different ways. Standard deviation uses all observations and is sensitive to outliers. IQR uses only the middle 50% and is resistant to outliers. Report SD when data are symmetric; consider IQR when data are skewed or have outliers.

18.8 Summary

- A **histogram** shows the shape of the distribution of a numerical variable.
- Common shapes: symmetric (bell-shaped), right-skewed (long right tail), left-skewed (long left tail).
- A **boxplot** displays Q1, median, Q3, whiskers, and outliers - a compact 5-number summary.
- The **IQR** is the range of the middle 50% of data; it is resistant to outliers.
- In Excel: use Insert → Statistical Chart for both histograms and boxplots.
- In R: use `hist()` for histograms and `boxplot()` for boxplots.

18.9 Practice Problems

1. A forester measures the diameter at breast height (DBH, in cm) of 10 trees in a stand: 12, 8, 35, 14, 11, 9, 42, 13, 10, 16. (a) Create a frequency table with bins of width 10 (0–9, 10–19, 20–29, 30–39, 40–49). (b) Calculate Q1, Q2, Q3, and IQR. (c) Are there any outliers according to the $1.5 \times$ IQR rule?

2. A researcher counts the number of invasive plant species found in 8 forest plots: 2, 1, 3, 5, 0, 2, 4, 1. Describe what you would expect the histogram to look like. Would you expect this distribution to be symmetric or skewed? Explain your reasoning.
3. Two ranches record their daily milk production (liters/cow). Ranch A: mean = 22 L, median = 21 L. Ranch B: mean = 22 L, median = 19 L. Without seeing the raw data, what do these statistics suggest about the shape of the distributions at each ranch? Which ranch likely has a few unusually high-producing cows?

19 Week 4 - Uncertainty and Variability

Dates: Wednesday September 9 · Friday September 11

(Note: Monday September 7 is Labor Day - no class)

19.1 Learning Objectives

By the end of this week, you should be able to:

- Explain the difference between a population and a sample.
- Define and calculate the standard error of the mean.
- Construct and interpret a 95% confidence interval using the rule of thumb.
- Explain why larger sample sizes produce more precise estimates.
- Recognize that random sampling introduces variability in our estimates.

19.2 Background and Conceptual Introduction

Suppose you want to know the average daily milk production of all dairy cows in Tennessee - that is the **population**. You cannot measure every cow in the state, so you measure a **sample** of 40 cows. You calculate the sample mean: 24 liters per day.

But here is the key question: how confident are you that the true population mean is close to 24? If you had sampled a different set of 40 cows, you would likely get a slightly different mean - maybe 23.5 or 24.8. This variation from sample to sample is called **sampling variability**, and it is at the heart of statistical inference.

Think of it like casting a fishing net. Each time you cast your net, you catch a different mix of fish. The average size of fish in your net is a good *estimate* of the average size of all fish in the lake - but it will not be perfect every time. How close your estimate is to the truth depends on the size of your net (sample size) and how variable the fish sizes are.

The **standard error** and **confidence interval** are tools that quantify our uncertainty. They tell us how far our sample estimate might be from the true population value.

Think About It

You sample 5 trees from a large forest and estimate the average diameter at breast height (DBH) to be 22 cm. Your classmate samples 50 trees from the same forest and estimates the average DBH to be 20 cm. Whose estimate is likely to be closer to the true population average? Why?

19.3 Key Concepts and Definitions

Population: The entire group of individuals or objects that you want to study. In practice, it is usually too large to measure completely.

Sample: A subset of the population that you actually measure.

Parameter: A numerical summary of a population (e.g., population mean μ). Usually unknown.

Statistic: A numerical summary calculated from a sample (e.g., sample mean $\bar{y}$). Used to estimate the parameter.

Sampling variability: The fact that different random samples from the same population will give slightly different statistics.

Standard Error of the Mean (SE): A measure of how much the sample mean would vary across repeated samples. Calculated as:

$$SE = \frac{s}{\sqrt{n}}$$

where s is the sample standard deviation and n is the sample size. A small SE means our estimate is precise.

Confidence Interval (CI): A range of values that is likely to contain the true population mean. A 95% CI means that if we repeated the sampling process many times, about 95% of the intervals we constructed would contain the true mean.

$$95\% \text{ CI} \approx \bar{y} \pm 2 \times SE$$

(This "rule of thumb" uses 2 as an approximation; the exact value depends on sample size and will be refined when we cover t-tests in Week 10.)

Precision vs. Accuracy: - **Precise** estimate: small standard error, values close together. - **Accurate** estimate: close to the true population value (requires no bias in sampling).

19.4 How to Do It by Hand

> Example: American Eel Length
>
> A fisheries biologist is studying American eels (*Anguilla rostrata*) in a New England pond. The pond has exactly 100 eels, and their true mean total length is known to be 100 cm. The biologist takes a random sample of 12 eels and records their lengths (cm):
> **Sample data:** 95, 102, 88, 110, 97, 104, 91, 99, 107, 93, 101, 96

Step 1: Calculate the sample mean

$$\bar{y} = \frac{95 + 102 + 88 + 110 + 97 + 104 + 91 + 99 + 107 + 93 + 101 + 96}{12} = \frac{1183}{12} = 98.6 \text{ cm}$$

Step 2: Calculate the sample standard deviation

First, find each deviation from the mean ($\bar{y} = 98.6$):

y_i	$y_i - \bar{y}$	$(y_i - \bar{y})^2$
95	-3.6	12.96
102	3.4	11.56
88	-10.6	112.36
110	11.4	129.96
97	-1.6	2.56
104	5.4	29.16
91	-7.6	57.76
99	0.4	0.16
107	8.4	70.56
93	-5.6	31.36
101	2.4	5.76
96	-2.6	6.76
Sum	0.0	**470.92**

$$s = \sqrt{\frac{470.92}{12-1}} = \sqrt{\frac{470.92}{11}} = \sqrt{42.81} = 6.54 \text{ cm}$$

Step 3: Calculate the standard error

$$SE = \frac{s}{\sqrt{n}} = \frac{6.54}{\sqrt{12}} = \frac{6.54}{3.46} = 1.89 \text{ cm}$$

Step 4: Calculate the 95% confidence interval

$$95\% \text{ CI} = \bar{y} \pm 2 \times SE = 98.6 \pm 2 \times 1.89 = 98.6 \pm 3.78$$

$$\text{Lower bound} = 98.6 - 3.78 = 94.8 \text{ cm}$$
$$\text{Upper bound} = 98.6 + 3.78 = 102.4 \text{ cm}$$

Interpretation: Based on our sample of 12 eels, we estimate that the true mean eel length is between 94.8 and 102.4 cm. The true mean (100 cm) falls inside our confidence interval - our sample gave a good estimate!

19.5 How to Do It in Excel

1. Enter your 12 eel lengths in column A (cells A1:A12).
2. In separate cells, calculate:

Cell	Formula	Result
B1	`=AVERAGE(A1:A12)`	Mean
B2	`=STDEV.S(A1:A12)`	Standard deviation
B3	`=COUNT(A1:A12)`	Sample size n
B4	`=B2/SQRT(B3)`	Standard error
B5	`=B1 - 2*B4`	Lower bound of 95% CI
B6	`=B1 + 2*B4`	Upper bound of 95% CI

> **Note**
>
> Screenshot will be added here.

19.6 How to Do It in R

```
# Eel length sample
eels <- c(95, 102, 88, 110, 97, 104, 91, 99, 107, 93, 101, 96)

# Sample mean
y_bar <- mean(eels)
cat("Mean:", y_bar, "\n")
```

```
Mean: 98.58333
```

```
# Sample standard deviation
s <- sd(eels)
cat("Std. deviation:", round(s, 2), "\n")
```

```
Std. deviation: 6.54
```

```
# Sample size
n <- length(eels)

# Standard error
SE <- s / sqrt(n)
cat("Standard error:", round(SE, 2), "\n")
```

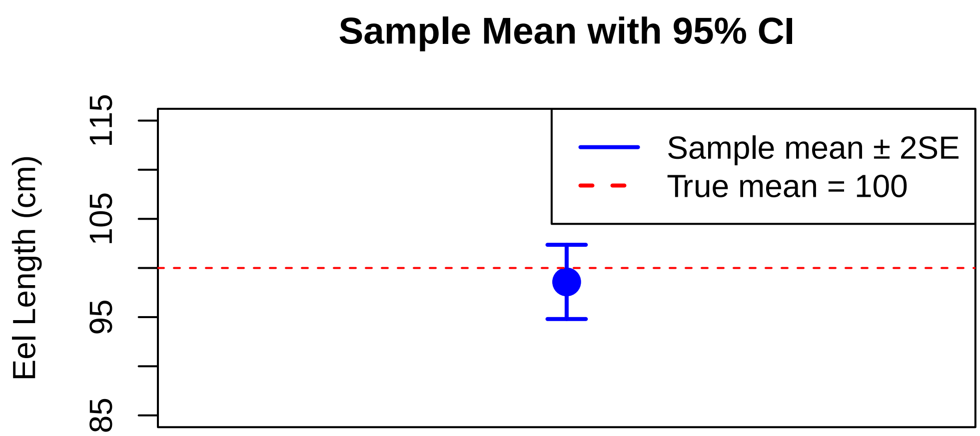
```
Standard error: 1.89
```

```
# 95% confidence interval (rule of thumb: mean ± 2*SE)
lower <- y_bar - 2 * SE
upper <- y_bar + 2 * SE
cat("95% CI (rule of thumb):", round(lower, 1), "to", round(upper, 1), "\n")
```

```
95% CI (rule of thumb): 94.8 to 102.4
```

```
# Visualize with a simple plot
# The true mean (100 cm) is shown as a reference
plot(x = 1, y = y_bar,
     xlim = c(0.5, 1.5), ylim = c(85, 115),
     pch = 16, cex = 2, col = "blue",
```

```r
     xaxt = "n",
     xlab = "", ylab = "Eel Length (cm)",
     main = "Sample Mean with 95% CI")
arrows(x0 = 1, y0 = lower, x1 = 1, y1 = upper,
       angle = 90, code = 3, length = 0.1, col = "blue", lwd = 2)
abline(h = 100, col = "red", lty = 2) # true population mean
legend("topright", legend = c("Sample mean ± 2SE", "True mean = 100"),
       col = c("blue", "red"), lty = c(1, 2), lwd = 2)
```



19.7 Interpretation and Common Mistakes

Effect of sample size on the standard error:

Notice that $SE = s/\sqrt{n}$. As n gets larger, the standard error gets smaller. This makes intuitive sense: a larger sample gives you a more precise estimate of the population mean.

n	s	SE
5	6.54	2.92

n	s	SE
12	6.54	1.89
50	6.54	0.92
100	6.54	0.65

For the same level of variability, tripling the sample size roughly cuts the SE in half.

> Common Mistake: Confusing Standard Deviation and Standard Error
>
> The **standard deviation (SD)** describes how much individuals in your sample vary from one another.
> The **standard error (SE)** describes how precisely your sample mean estimates the population mean.
> The SE is always smaller than the SD (for $n > 1$). Reporting the SE as a measure of data spread is misleading - it understates the true variability among individuals.

> Common Mistake: Misinterpreting a 95% CI
>
> A 95% confidence interval does NOT mean "there is a 95% chance the true mean is in this interval." Once you have calculated an interval, the true mean is either in it or it is not. The 95% refers to the procedure: if you repeated the sampling 100 times and built 100 CIs, about 95 of them would contain the true mean. This distinction will matter more when we get to hypothesis testing.

19.8 Summary

- A **sample** is used to estimate a **population** parameter when we cannot measure everyone.
- **Sampling variability** means different samples give different estimates.
- The **standard error** measures how much the sample mean varies from sample to sample: $SE = s/\sqrt{n}$.
- A **95% confidence interval** ($\bar{y} \pm 2 \times SE$) gives a range of plausible values for the true population mean.
- Larger sample sizes reduce the standard error and produce narrower confidence intervals.

19.9 Practice Problems

1. A wildlife manager samples 9 wild turkeys and records their weights (kg): 6.2, 5.8, 7.1, 6.5, 5.9, 6.8, 7.0, 6.3, 6.7. Calculate the mean, standard deviation, standard error, and 95% confidence interval. Show all steps.

2. A researcher estimates the average phosphorus concentration in a stream using two studies: Study A used $n = 10$ samples (mean = 0.45 mg/L, SD = 0.12) and Study B used $n = 80$ samples (mean = 0.42 mg/L, SD = 0.11). Calculate the standard error for each study. Which study provides a more precise estimate of the true mean? Why?

3. If you quadrupled the sample size (from n to $4n$), by what factor would the standard error decrease? Show this using the formula.

Part III

Unit 2: Probability and Uncertainty

20 Week 5 - Frequency Data and Probability

Dates: Monday September 14 · Wednesday September 16 · Friday September 18

20.1 Learning Objectives

By the end of this week, you should be able to:

- Define probability and explain the difference between empirical and theoretical probability.
- Construct a frequency table and a relative frequency table from a dataset.
- Describe the properties of a normal distribution.
- Use the empirical rule (68–95–99.7 rule) to answer basic probability questions.
- Calculate a z-score and explain what it means.

20.2 Background and Conceptual Introduction

Probability is the foundation of all statistical inference. Before we can test whether a fertilizer treatment increases yield or whether a fish population is declining, we need to understand what "by chance alone" looks like.

Think about rolling a fair six-sided die. You know ahead of time that the probability of rolling a 4 is 1/6 16.7%. That is a *theoretical probability* - derived from reasoning about the die. Now imagine you actually rolled the die 60 times and got a 4 on 12 of those rolls (12/60 = 20%). That is an *empirical probability* - derived from observed data. The two values differ slightly because of random chance in the 60 trials.

In agriculture and natural resources, we almost always work with empirical probabilities derived from data. For example:

- In 10 years of records, late spring frosts occurred in 3 of those years → estimated probability of a late frost = 3/10 = 30%.
- In a survey of 200 prairie plots, 74 were occupied by the target bird species → estimated probability of occupancy = 74/200 = 37%.

Think About It

A seed company claims their new soybean variety germinates 90% of the time under field conditions. You plant 20 seeds and only 14 germinate (70%). Does this mean the company is lying? Or could a 70% germination rate happen by random chance when the true rate is 90%? How would you decide?

20.3 Key Concepts and Definitions

Probability: A number between 0 and 1 that represents how likely an event is. Probability = 0 means impossible; probability = 1 means certain.

Empirical probability: Estimated from observed data - the proportion of times an event occurred in past trials.

$$P(\text{event}) = \frac{\text{number of times event occurred}}{\text{total number of trials}}$$

Frequency table: A table showing how many observations fall in each category or value.

Relative frequency: The proportion (or percentage) of observations in each category. Relative frequency = frequency / total n.

Random variable: A variable whose value is determined by a random process (e.g., the weight of a randomly selected deer from a population).

Probability distribution: A description of all possible values of a random variable and their associated probabilities.

Normal distribution: A symmetric, bell-shaped probability distribution completely described by its mean (μ) and standard deviation (σ). It appears throughout nature - heights, weights, measurement errors, and many other biological quantities follow approximately normal distributions.

The Empirical Rule (68–95–99.7 Rule): For data that follow a normal distribution: - About **68%** of observations fall within 1 standard deviation of the mean ($\mu \pm 1\sigma$). - About **95%** of observations fall within 2 standard deviations of the mean ($\mu \pm 2\sigma$). - About **99.7%** of observations fall within 3 standard deviations of the mean ($\mu \pm 3\sigma$).

z-score (standard score): The number of standard deviations an observation is above or below the mean:

$$z = \frac{y - \mu}{\sigma}$$

A z-score of 0 means the observation is exactly at the mean. A z-score of +2 means the observation is 2 standard deviations above the mean.

20.4 How to Do It by Hand

> **Example: Germination Counts**
>
> A plant scientist records the number of seeds (out of 10) that germinated in 15 pots treated with a new soil amendment:
> **Data:** 7, 9, 8, 10, 6, 9, 8, 7, 10, 8, 9, 6, 8, 9, 7

20.4.1 Frequency Table

Step 1: Count occurrences of each value

Germination count	Frequency	Relative Frequency
6	2	$2/15 = 0.133$
7	3	$3/15 = 0.200$
8	4	$4/15 = 0.267$
9	4	$4/15 = 0.267$
10	2	$2/15 = 0.133$
Total	**15**	**1.000**

The most common germination count is 8 or 9 seeds out of 10 (about 27% each). The data center around 8, which suggests roughly 80% germination per pot.

20.4.2 Using the Empirical Rule

Suppose the weight of mature corn ears at a research farm follows a normal distribution with mean $\mu = 185$ g and standard deviation $\sigma = 15$ g.

What proportion of ears weigh between 170 g and 200 g?

$170 = 185 - 15 = \mu - 1\sigma$ $200 = 185 + 15 = \mu + 1\sigma$

By the empirical rule, about **68%** of ears weigh between 170 and 200 g.

What proportion of ears weigh between 155 g and 215 g?

$155 = 185 - 30 = \mu - 2\sigma$ $215 = 185 + 30 = \mu + 2\sigma$

By the empirical rule, about **95%** of ears fall in this range.

What is the z-score for an ear weighing 220 g?

$$z = \frac{220 - 185}{15} = \frac{35}{15} = 2.33$$

This ear is 2.33 standard deviations above the mean - it is quite large but not impossible.

20.5 How to Do It in Excel

20.5.1 Frequency Table

1. Enter your raw data in column A.
2. In column C, list all unique values (the "bins").
3. Use `=COUNTIF(A:A, C1)` in column D to count how many times each value appears.
4. Calculate relative frequency in column E: `=D1/SUM(D:D)`.

> **Note**
>
> Screenshot will be added here.

20.5.2 Normal Distribution Probabilities

Excel has built-in functions for the normal distribution:

- `=NORM.DIST(x, mean, std_dev, TRUE)` - gives the probability that a normal random variable is x (cumulative probability).
- `=NORM.DIST(200, 185, 15, TRUE)` - probability an ear weighs 200 g.
- `=NORM.DIST(200,185,15,TRUE) - NORM.DIST(170,185,15,TRUE)` - probability between 170 and 200 g.

20.6 How to Do It in R

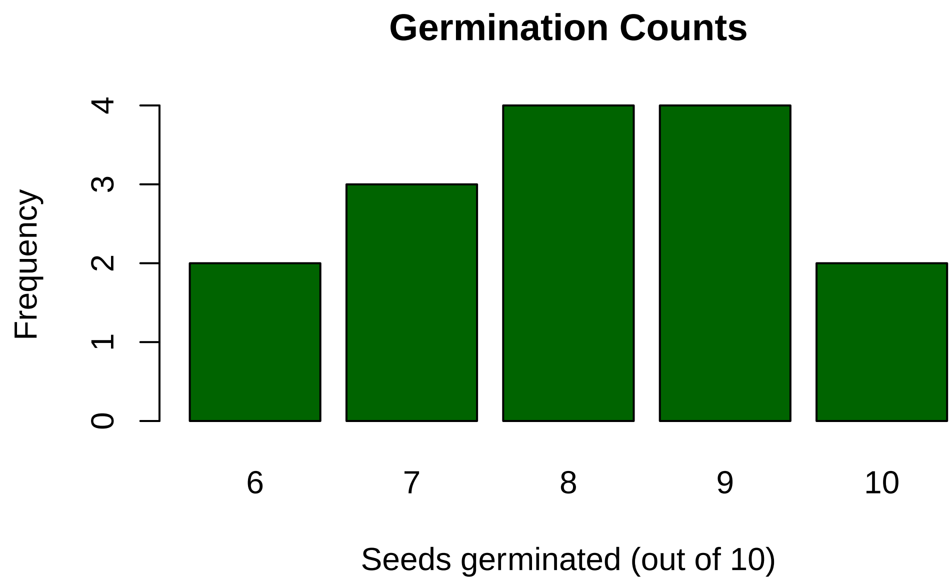
```r
# Germination data
germ <- c(7, 9, 8, 10, 6, 9, 8, 7, 10, 8, 9, 6, 8, 9, 7)

# Frequency table
table(germ)
```

```
germ
 6  7  8  9 10 
 2  3  4  4  2 
```

```r
# Relative frequency table
table(germ) / length(germ)
```

```
germ
        6         7         8         9        10 
0.1333333 0.2000000 0.2666667 0.2666667 0.1333333 
```

```r
# Bar plot of frequencies
barplot(table(germ),
        main = "Germination Counts",
        xlab = "Seeds germinated (out of 10)",
        ylab = "Frequency",
        col  = "darkgreen")
```




```r
# --- Normal distribution examples ---
# Corn ear weight: mean = 185, sd = 15

# P(weight <= 200) using normal distribution
pnorm(200, mean = 185, sd = 15)
```

```
[1] 0.8413447
```

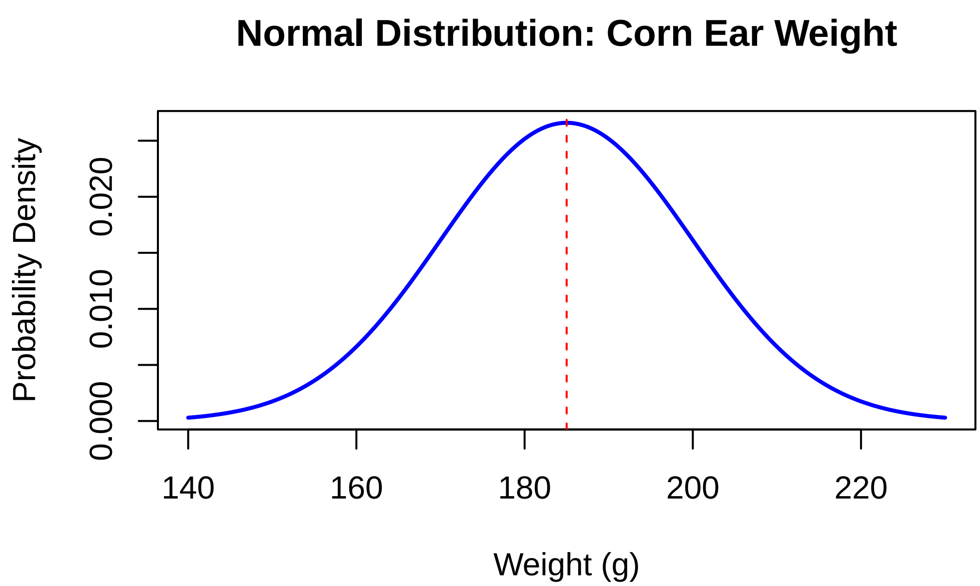
```r
# P(170 <= weight <= 200)
pnorm(200, mean = 185, sd = 15) - pnorm(170, mean = 185, sd = 15)
```

```
[1] 0.6826895
```

```r
# z-score for a 220 g ear
z <- (220 - 185) / 15
cat("z-score for 220 g:", z, "\n")
```

```
z-score for 220 g: 2.333333
```

```r
# Plot a normal distribution curve
x_vals <- seq(140, 230, length.out = 300)
y_vals <- dnorm(x_vals, mean = 185, sd = 15)

plot(x_vals, y_vals, type = "l", lwd = 2, col = "blue",
     main = "Normal Distribution: Corn Ear Weight",
     xlab = "Weight (g)", ylab = "Probability Density")
abline(v = 185, col = "red", lty = 2) # mean
```



20.7 Interpretation and Common Mistakes

What does probability mean in practice?

A probability of 0.05 (5%) does not mean something is "impossible" - it means it happens about 1 time in 20. This number (5%) plays a central role in hypothesis testing, which we will cover in Weeks 8–9.

Common Mistake: Applying the Empirical Rule to Non-Normal Data

The 68–95–99.7 rule only applies to data that follow a normal distribution. If your data are strongly skewed or have many outliers, the empirical rule will give wrong answers. Always look at a histogram first to assess whether data appear roughly normal.

Common Mistake: Confusing Frequency and Probability

Frequency is a count (e.g., 4 pots had 8 germinating seeds). Probability is a proportion between 0 and 1 (e.g., $4/15 = 0.267$). Make sure you know which one is being asked for.

20.8 Summary

- **Probability** quantifies how likely an event is; it ranges from 0 to 1.
- **Frequency tables** organize count data; **relative frequency** tables show proportions.
- The **normal distribution** is bell-shaped, symmetric, and described by mean and standard deviation.
- The **empirical rule**: 68% of data within 1 SD, 95% within 2 SD, 99.7% within 3 SD.
- A **z-score** measures how many standard deviations an observation is from the mean.

20.9 Practice Problems

1. A wildlife technician checks 20 nest boxes and finds eggs in 8 of them. What is the empirical probability that a randomly selected nest box contains eggs? Express your answer as both a fraction and a decimal.

2. Corn grain moisture content at harvest at a particular farm follows a normal distribution with mean = 15% and standard deviation = 2%. (a) What proportion of samples are between 13% and 17%? (b) What proportion are above 19%? (c) Calculate the z-score for a sample with 11% moisture.

3. A researcher counts the number of earthworms in 12 soil cores: 3, 5, 2, 8, 4, 6, 3, 7, 5, 4, 6, 5. Construct a relative frequency table. What is the most common count? What proportion of cores had 5 or more earthworms?

21 Week 6 — Binomial Proportions

Dates: Monday September 21 · Wednesday September 23 · Friday September 25

21.1 Learning Objectives

By the end of this week, you should be able to:

- Identify when a situation follows a binomial distribution (the BINS conditions).
- Calculate binomial probabilities using the formula.
- Estimate a population proportion from a sample proportion.
- Construct a confidence interval for a proportion.
- Apply binomial reasoning to real agricultural and natural resource scenarios.

21.2 Background and Conceptual Introduction

Many questions in agriculture and natural resources involve *proportions* — not means or standard deviations, but the fraction of individuals in a category:

- What proportion of seedlings survive transplanting?
- What percentage of grain samples are contaminated with mycotoxins?
- What fraction of wildlife camera trap photos contain the target species?
- What proportion of fish in a sample carry a specific disease?

When we measure a proportion, we are dealing with a fundamentally different type of data than the continuous measurements we handled in Weeks 2–4. Each observation is a simple **yes/no** outcome: germinated or not, infected or not, detected or not. This type of data follows a **binomial distribution**.

Imagine flipping a coin 10 times. Each flip is independent — previous flips do not affect the next one. The outcome is always heads or tails. You can ask: "What is the probability of getting exactly 7 heads?" The binomial distribution answers this question.

Now replace "coin flip" with "does a newborn calf survive its first week?" or "does a planted seed germinate?" The math is exactly the same.

Think About It

A corn seed variety has a germination rate of 85%. If you plant 10 seeds, how many do you expect to germinate? Would it surprise you if only 6 germinated? What about if none germinated? The binomial distribution helps us judge which outcomes are plausible and which are extraordinary.

21.3 Key Concepts and Definitions

Bernoulli trial: A single experiment with exactly two possible outcomes — success or failure. The probability of success is p.

Binomial distribution: The distribution of the number of successes in n independent Bernoulli trials, each with the same probability of success p.

The BINS conditions (all four must be met): 1. **B**inary outcome — only two possible outcomes (success / failure) 2. **I**ndependent trials — the outcome of one trial does not affect others 3. **N**umber of trials is fixed in advance (n is set before you start) 4. **S**ame probability of success (p) on each trial

Binomial probability formula: The probability of getting exactly k successes in n trials is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is "n choose k" — the number of ways to arrange k successes among n trials.

Sample proportion ($\hat{p}$): The observed proportion of successes in a sample:

$$\hat{p} = \frac{\text{number of successes}}{n}$$

Standard error of a proportion:

$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

95% Confidence interval for a proportion (rule of thumb):

$$95\% \text{ CI} \approx \hat{p} \pm 2 \times SE_{\hat{p}}$$

21.4 How to Do It by Hand

> **Example: Calf Survival**
>
> On a beef cattle operation, the probability that a newborn calf survives its first 30 days is $p = 0.90$. In a group of 8 calves born this spring, what is the probability that exactly 7 survive?

Check BINS conditions: - Binary: calf survives or does not - Independent: survival of one calf does not affect others (assume so) - Fixed n: $n = 8$ calves - Same p: $p = 0.90$ for each calf

Calculate $\binom{8}{7}$:

$$\binom{8}{7} = \frac{8!}{7! \cdot 1!} = \frac{8 \times 7!}{7! \times 1} = 8$$

Apply the formula:

$$P(X = 7) = 8 \times (0.90)^7 \times (0.10)^1$$

$$= 8 \times 0.4783 \times 0.10 = 8 \times 0.04783 = 0.3826$$

There is about a **38% chance** that exactly 7 of the 8 calves survive.

> **Example: Estimating Camera Trap Detection Rate**
>
> A wildlife biologist checks 50 camera trap sites over a season. Deer were detected at 32 of the sites. Estimate the proportion of sites where deer are present, and construct a 95% confidence interval.

Sample proportion:

$$\hat{p} = \frac{32}{50} = 0.64 \quad (64\%)$$

Standard error:

$$SE_{\hat{p}} = \sqrt{\frac{0.64 \times 0.36}{50}} = \sqrt{\frac{0.2304}{50}} = \sqrt{0.004608} = 0.0679$$

95% CI:

$$0.64 \pm 2 \times 0.0679 = 0.64 \pm 0.136$$

$$\text{Lower: } 0.504 \quad \text{Upper: } 0.776$$

Interpretation: We estimate that deer are present at 64% of camera trap sites. We are 95% confident the true detection proportion is between 50.4% and 77.6%.

21.5 How to Do It in Excel

21.5.1 Binomial Probabilities

- **Exactly k successes:** `=BINOM.DIST(k, n, p, FALSE)`
  - Example: `=BINOM.DIST(7, 8, 0.9, FALSE)` → 0.383
- **At most k successes (cumulative):** `=BINOM.DIST(k, n, p, TRUE)`
  - Example: `=BINOM.DIST(7, 8, 0.9, TRUE)` → probability of 7 or fewer surviving

21.5.2 CI for a Proportion

Cell	Formula	Description
B1	`=32/50`	Sample proportion $\hat{p}$
B2	`=SQRT(B1*(1-B1)/50)`	Standard error
B3	`=B1 - 2*B2`	Lower CI bound
B4	`=B1 + 2*B2`	Upper CI bound

> **Note**
>
> Screenshot will be added here.

21.6 How to Do It in R

```r
# Binomial probability: exactly 7 of 8 calves survive
# P(X = 7), n=8, p=0.90
dbinom(7, size = 8, prob = 0.90)
```

```
[1] 0.3826375
```

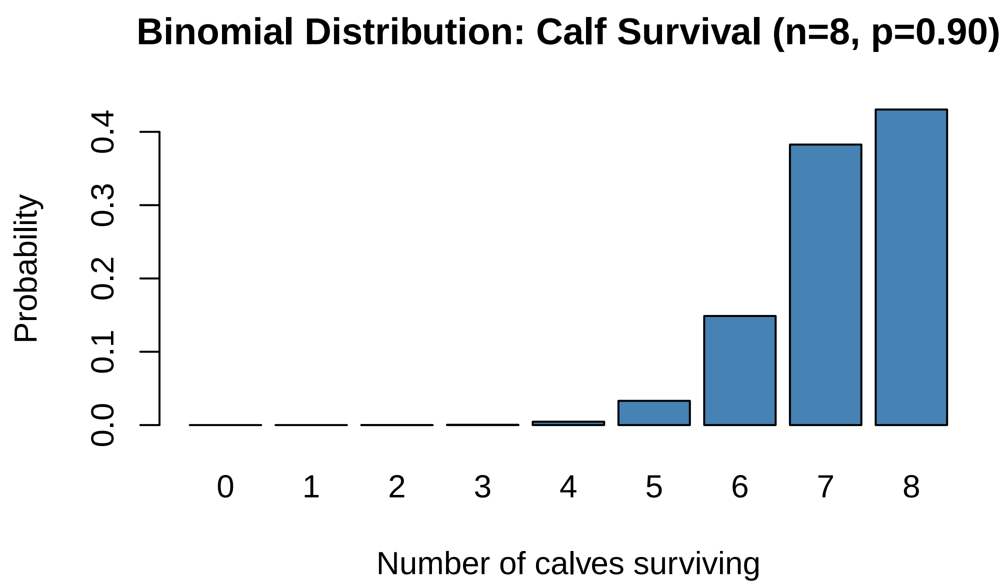
```r
# Cumulative: P(X <= 7)
pbinom(7, size = 8, prob = 0.90)
```

```
[1] 0.5695328
```

```r
# Probability of 7 OR 8 calves surviving (at least 7)
pbinom(6, size = 8, prob = 0.90, lower.tail = FALSE) # P(X > 6) = P(X >= 7)
```

```
[1] 0.8131047
```

```r
# --- Plotting a binomial distribution ---
k_vals <- 0:8
probs  <- dbinom(k_vals, size = 8, prob = 0.90)

barplot(probs, names.arg = k_vals,
        main = "Binomial Distribution: Calf Survival (n=8, p=0.90)",
        xlab = "Number of calves surviving",
        ylab = "Probability",
        col  = "steelblue")
```




```r
# --- CI for a proportion ---
n_sites    <- 50
n_detected <- 32
p_hat      <- n_detected / n_sites
SE_p       <- sqrt(p_hat * (1 - p_hat) / n_sites)

cat("Sample proportion:", p_hat, "\n")
```

```
Sample proportion: 0.64
```

```r
cat("Standard error:", round(SE_p, 4), "\n")
```

```
Standard error: 0.0679
```

```r
cat("95% CI:", round(p_hat - 2*SE_p, 3), "to", round(p_hat + 2*SE_p, 3), "\n")
```

```
95% CI: 0.504 to 0.776
```

```r
# Built-in exact test for proportions
prop.test(n_detected, n_sites, conf.level = 0.95)
```

```

	1-sample proportions test with continuity correction

data:  n_detected out of n_sites, null probability 0.5
X-squared = 3.38, df = 1, p-value = 0.06599
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.4914300 0.7671498
sample estimates:
   p 
0.64 
```

21.7 Interpretation and Common Mistakes

Expected value and variance of a binomial:

If X follows a binomial distribution with parameters n and p: - Expected number of successes: $E(X) = np$ - Standard deviation: $\sigma_X = \sqrt{np(1-p)}$

For our calf example: $E(X) = 8 \times 0.90 = 7.2$ calves survive on average; $\sigma = \sqrt{8 \times 0.90 \times 0.10} = \sqrt{0.72} = 0.85$.

> ⚠ Common Mistake: Applying Binomial to Non-Independent Trials
>
> The binomial requires independent trials. If calves are from the same mother (twins) or share a disease exposure, survival is not independent and the binomial is not valid.

Always check the independence assumption.

⚠ Common Mistake: Using Binomial When n or p is Not Fixed

If you count deer until you find 10 (rather than checking a fixed number of sites), the number of trials is not fixed — this is a *negative binomial* situation, not binomial. Always check whether n is set before the experiment begins.

⚠ Common Mistake: Wide CI from Small Samples

With $n = 10$ camera sites, even observing detection at 6 of 10 gives a very wide CI (approximately 0.30 to 0.90). Small samples give unreliable proportion estimates. You generally need $n \geq 30$ for a reasonably precise CI.

21.8 Summary

- A **binomial distribution** models the number of successes in n independent trials with success probability p.
- The **BINS conditions** must all hold: binary outcomes, independent trials, fixed n, same p.
- Use $\hat{p} = k/n$ to estimate a population proportion from a sample.
- The **standard error** of a proportion is $\sqrt{\hat{p}(1-\hat{p})/n}$.
- The **95% CI** for a proportion is approximately $\hat{p} \pm 2 \times SE_{\hat{p}}$.

21.9 Practice Problems

1. A forestry experiment plants 12 tree seedlings in a degraded area. Based on past experience, each seedling has a 70% probability of establishing successfully (independent of the others). (a) What is the probability that exactly 10 seedlings survive? Show your work using the binomial formula. (b) What is the expected number of surviving seedlings?

2. A fish hatchery samples 40 fingerlings and finds that 28 test positive for a bacterial infection. (a) Calculate the sample proportion. (b) Calculate the standard error. (c) Construct a 95% confidence interval. (d) Interpret the confidence interval in one plain-language sentence.

3. Does the following scenario meet the BINS conditions for a binomial distribution? Explain. "A farmer randomly plants seeds from a bag and records how many of the first 20 planted germinate, but stops counting if 3 consecutive seeds fail to germinate."

22 Week 7 — Experimental Design and the Scientific Method

Dates: Monday September 28 · Wednesday September 30 · Friday October 2

22.1 Learning Objectives

By the end of this week, you should be able to:

- Explain the steps of the scientific method and how they apply to agricultural and natural resource research.
- Define treatment, control, replication, randomization, and blocking.
- Distinguish between observational studies and controlled experiments.
- Identify potential sources of bias in an experimental design.
- Design a basic experiment for a simple agricultural or natural resource question.

22.2 Background and Conceptual Introduction

You have spent the first six weeks of this course learning *how* to describe and summarize data. Now it is time to step back and ask: *where does data come from, and how should we collect it?*

The way you design an experiment determines whether you can draw valid conclusions. No amount of sophisticated statistics can rescue a poorly designed study. The old saying in research is: "Design your experiment so that the statistics you need are obvious." In other words, if you have to ask how to analyze your data, you may need to redesign your study.

Consider this scenario: A farmer applies a new herbicide to his fields and notices that weed pressure drops significantly that year. Does the herbicide work? Maybe — but maybe it was also a drier-than-normal year that reduced weed germination. Or maybe the treated fields were already less weedy to begin with. Without proper experimental design, we cannot separate the effect of the herbicide from these other potential explanations. These alternative explanations are called **confounding variables**.

Good experimental design is like building a fair test. Scientists who study crop varieties want to know: "Is this new variety truly better, or did it just get lucky with a good plot of land?" The three pillars of good experimental design — **randomization**, **replication**, and **controls** — are designed specifically to eliminate luck and confounding.

> Think About It
>
> Imagine you want to test whether playing classical music in a greenhouse increases tomato plant growth. You play music in one greenhouse (your "treatment") and leave the other greenhouse silent (your "control"). After 4 weeks, the music greenhouse plants are taller. But the music greenhouse also happened to get more sunlight because it faces south. How does this flaw in the design undermine your conclusion? How would you fix it?

22.3 Key Concepts and Definitions

Scientific method: The systematic process of asking a question, forming a hypothesis, designing an experiment, collecting data, analyzing results, and drawing conclusions.

Hypothesis: A testable prediction about the relationship between variables. Must be falsifiable.

Experimental unit: The individual entity that receives a treatment (e.g., one field plot, one cow, one seedling).

Treatment: What you apply or manipulate in the experiment (e.g., fertilizer type, irrigation level, feed supplement).

Control group: A group that receives no treatment (or the standard/reference treatment). The control provides a baseline for comparison.

Replication: Using multiple experimental units per treatment. Replication is essential — it lets you estimate natural variability and separate real treatment effects from random noise.

Randomization: Randomly assigning experimental units to treatments. Randomization distributes unknown confounding variables evenly across groups.

Blocking: Grouping similar experimental units together (a "block") before randomizing treatments within each block. Blocking controls for known sources of variability (e.g., soil type, field slope).

Confounding variable: A variable that is related to both the treatment and the outcome, making it hard to isolate the treatment effect.

Observational study: A study where the researcher measures but does not manipulate variables. Can identify associations but cannot establish cause and effect.

Controlled experiment: A study where the researcher actively assigns treatments to experimental units. When properly designed, allows causal conclusions.

22.4 How to Design an Experiment

> **Example: Testing a New Fertilizer**
>
> A crop scientist wants to know whether a new slow-release nitrogen fertilizer increases corn yield compared to the standard fertilizer.
> **Research question:** Does slow-release nitrogen fertilizer increase corn yield?
> **Hypothesis:** Corn plots receiving slow-release nitrogen will have higher yield (bushels/acre) than plots receiving standard nitrogen fertilizer.

Step 1: Identify the response variable

Corn grain yield (bushels/acre) — this is what we measure to assess the treatment effect.

Step 2: Identify the explanatory variable (treatment)

Fertilizer type: standard nitrogen vs. slow-release nitrogen.

Step 3: Decide on a control

The standard fertilizer serves as the control — it is the current practice.

Step 4: Determine the experimental units

Each experimental unit is one 1-acre plot. We will use 20 plots total.

Step 5: Randomize

Randomly assign 10 plots to receive standard fertilizer and 10 plots to receive slow-release fertilizer. To do this, number the plots 1–20 and use a random number generator (Excel: `=RAND()`, or draw numbers from a hat).

Step 6: Consider blocking

If the field has a noticeable slope (wet low end vs. dry high end), we divide the field into blocks of 2 plots each — one wet and one dry. Within each block, randomly assign one plot to each treatment. This ensures both fertilizer types experience similar water conditions.

Step 7: Identify potential confounders and minimize them

Apply both fertilizers at the same rate (kg N/acre). Plant the same variety. Use the same planting date. The only thing that differs between groups is the fertilizer type.

Step 8: Collect data consistently

Harvest all plots at the same time, with the same equipment, by the same operators.

22.5 Types of Study Designs

Design Type	Can Manipulate?	Controls?	Randomization?	Can Infer Causation?
Observational	No	No	No	No (only association)
Quasi-experiment	Sometimes	Partial	No	Weak
Randomized controlled experiment	Yes	Yes	Yes	Yes
Randomized complete block design	Yes	Yes	Yes (within block)	Yes

---

22.6 How to Do It in R

```r
# Simulating random assignment of 20 plots to two treatments
set.seed(42) # set seed for reproducibility

plots     <- 1:20
treatment <- sample(rep(c("Standard", "Slow-release"), each = 10))

assignment <- data.frame(Plot = plots, Treatment = treatment)
print(assignment)
```

```
   Plot    Treatment
1     1 Slow-release
2     2     Standard
3     3     Standard
4     4     Standard
5     5     Standard
6     6     Standard
7     7 Slow-release
8     8 Slow-release
9     9     Standard
10   10     Standard
11   11 Slow-release
12   12     Standard
13   13 Slow-release
14   14     Standard
15   15 Slow-release
16   16 Slow-release
17   17 Slow-release
18   18     Standard
19   19 Slow-release
20   20 Slow-release
```

```r
# Check the balance
table(assignment$Treatment) # should be 10 in each group
```

```

Slow-release     Standard 
          10           10 
```

22.7 How to Spot a Poorly Designed Study

> **Common Mistake: No Replication (n = 1)**
>
> A farmer applies a new herbicide to one field and the old herbicide to a different field, then compares weed counts. There is no replication — each treatment is applied to only one field. Any difference could be due to the herbicide or to inherent differences between the two fields. You cannot draw conclusions from a single observation per treatment.

> **Common Mistake: Confusing Correlation with Causation**
>
> An observational study finds that farms with more land under cover crops have higher profit margins. Does planting cover crops *cause* higher profits? Not necessarily — larger, more sophisticated farms may be both more likely to plant cover crops AND more profitable for other reasons. Only a randomized experiment can establish causation.

> **Common Mistake: Pseudoreplication**
>
> Using multiple measurements from the same experimental unit as if they were independent replicates. For example: measuring yield at 5 spots within a single fertilized plot and treating those 5 measurements as 5 independent replicates. They are not — they all came from the same unit receiving the same treatment.

22.8 Interpretation and Common Mistakes

Why does randomization matter?

Randomization is the key step that allows causal inference. By randomly assigning treatments, we ensure that any difference we observe between groups is not due to systematic bias — it is either a real treatment effect or random chance. Statistical tests (coming in Weeks 8–9) then help us decide which explanation is more likely.

Why does replication matter?

Individual plants, animals, and fields vary naturally. With only one or two replicates per treatment, we cannot tell whether a difference between groups is real or just due to the natural variation between individual units. More replicates → more information → more reliable conclusions.

22.9 Summary

- Good experimental design requires **randomization**, **replication**, and a **control group**.
- **Randomization** prevents confounding; **replication** allows estimation of random variability.
- **Blocking** controls for known sources of variability (e.g., field topography, soil type).

- An **observational study** can show association but cannot prove causation.
- A well-designed **randomized controlled experiment** is the gold standard for establishing cause and effect.
- Common design flaws: no replication, no randomization, pseudoreplication, and uncontrolled confounders.

22.10 Practice Problems

1. A wildlife manager wants to test whether supplemental feeding increases white-tailed deer body mass in winter. She places feeding stations in 6 of the 12 forest blocks in her management area and leaves the other 6 blocks as controls. She weighs a sample of deer from each block at the end of winter. (a) What is the experimental unit? (b) What is the response variable? (c) Is there replication? (d) How could she randomize the assignment of feeding stations to blocks?

2. A researcher reports that regions with higher livestock density have higher rates of stream phosphorus contamination. Is this an observational study or a controlled experiment? Can the researcher conclude that livestock *cause* phosphorus contamination? What alternative explanations exist?

3. Design a simple experiment to test whether adding compost to garden soil increases tomato yield. Specify: research question, hypothesis, experimental units, treatments, control, replication (how many units per treatment?), and one potential confounding variable you would need to control.

Part IV

Unit 3: Inference and Hypothesis Testing

23 Week 8 — Hypothesis Testing: Conceptual Introduction

Dates: Wednesday October 7 · Friday October 9

(Note: Monday October 5 — no class)

23.1 Learning Objectives

By the end of this week, you should be able to:

- State null and alternative hypotheses for a given research question.
- Explain what a p-value means in plain language.
- Define the significance level (α) and explain its role in decision-making.
- Distinguish between Type I and Type II errors.
- Apply the four-step NHST framework to a conceptual example.

23.2 Background and Conceptual Introduction

You now know how to describe data (Units 1 and 2) and how to collect it properly (Week 7). The next challenge is this: once you have data from a well-designed experiment, how do you decide whether an observed difference is *real* or just due to *random chance*?

This is the core problem of statistical inference, and the solution is **hypothesis testing**. Hypothesis testing is a formal decision-making framework that lets you judge, based on data, whether the evidence supports a claim.

Here is an analogy. In the United States legal system, a defendant is presumed innocent until proven guilty. The prosecution must present enough evidence to convince a jury beyond a reasonable doubt. Notice three things:

1. We start with a *default assumption* (innocent).
2. We set a *standard of evidence* (beyond reasonable doubt).

3. We make a *decision* (guilty or not guilty) based on the evidence.

Statistical hypothesis testing works the same way:

1. We start with a *default assumption* called the **null hypothesis** (H_0).
2. We set a *standard of evidence* called the **significance level** (α, usually 0.05).
3. We make a *decision* (reject H_0 or fail to reject H_0) based on a calculated **p-value**.

> **Think About It**
>
> A new insecticide is claimed to reduce aphid counts on wheat by 50%. You apply it to 15 wheat plots and count aphids before and after. You find a 35% reduction. Is 35% close enough to 50% to support the claim? Or could a 35% reduction happen by random chance even if the true effect is 50%? How would you decide?

23.3 Key Concepts and Definitions

Null hypothesis (H_0): The default assumption — usually "no effect," "no difference," or "no relationship." We test against this assumption. Written in terms of population parameters.

Alternative hypothesis (H_a or H_1): The research hypothesis — what we are trying to find evidence for. Written as the complement of H_0.

Test statistic: A number calculated from the data that measures how far the observed result is from what H_0 predicts. Large test statistics (far from zero) provide more evidence against H_0.

p-value: The probability of observing a result as extreme as (or more extreme than) the one you got, *assuming H_0 is true.* A small p-value means the data are surprising under H_0.

Significance level (α): The threshold below which we reject H_0. The most common value is $\alpha = 0.05$ (5%). We set α before collecting data.

Decision rule: - If $p \leq \alpha$: **Reject** H_0 — the result is "statistically significant" - If $p > \alpha$: **Fail to reject** H_0 — insufficient evidence to reject

Type I Error (α): Rejecting H_0 when it is actually true — a "false positive." We convict an innocent person.

Type II Error (β): Failing to reject H_0 when it is actually false — a "false negative." We acquit a guilty person.

Decision	H_0 true	H_0 false
Reject H_0	**Type I Error** (α)	Correct (Power)
Fail to reject H_0	Correct	**Type II Error** (β)

23.4 The Four-Step NHST Framework

We will use these four steps for every hypothesis test in this course:

Step 1: State the hypotheses

Write H_0 and H_a clearly using proper notation. Be specific about the population parameter.

Step 2: Set α and select the test

Choose a significance level (usually 0.05). Choose the appropriate test statistic for your data type and study design.

Step 3: Calculate the test statistic and p-value

Use the data to calculate the test statistic. Determine the p-value (using a table, Excel, or R).

Step 4: Make a decision and interpret

Compare the p-value to α. State whether you reject or fail to reject H_0. Write a conclusion in the context of the original research question.

23.5 A Conceptual Walk-Through

Example: Average Chicken Feed Consumption

A chicken feed manufacturer claims that hens eat an average of 120 g of feed per day. A farm manager suspects her flock eats *more* than this. She randomly selects 25 hens and records their daily feed intake over one week.
She finds: $\bar{y} = 127$ g/day, $s = 18$ g.

Step 1: State the hypotheses

$$H_0 : \mu = 120 \text{ g/day} \quad \text{(manufacturer's claim)}$$

$$H_a : \mu > 120 \text{ g/day} \quad \text{(the manager's suspicion — one-sided test)}$$

Step 2: Set α

$\alpha = 0.05$. We will use a one-sample t-test (covered fully next week).

Step 3: Calculate the test statistic

$$t = \frac{\bar{y} - \mu_0}{SE} = \frac{127 - 120}{18/\sqrt{25}} = \frac{7}{3.6} = 1.94$$

The p-value for $t = 1.94$ with 24 degrees of freedom (one-tailed) is approximately 0.032.

Step 4: Decision and interpretation

$p = 0.032 < \alpha = 0.05 \rightarrow$ **Reject** H_0

Conclusion: There is sufficient evidence ($t_{24} = 1.94$, $p = 0.032$) to conclude that hens at this farm consume more than 120 g of feed per day on average.

23.6 Interpretation and Common Mistakes

! What a p-value is NOT

The p-value is frequently misunderstood. Be careful:

- **p-value is NOT** the probability that H_0 is true.
- **p-value is NOT** the probability that your result happened by chance.
- **p-value IS** the probability of observing data as extreme as yours IF H_0 were true.

These distinctions seem subtle but matter greatly when interpreting results.

⚠ Common Mistake: "Proving" the Null Hypothesis

When $p > \alpha$ and we "fail to reject H_0," we do **not** prove that H_0 is true. We simply lack sufficient evidence to reject it. A not-guilty verdict in a trial does not prove innocence — it just means the evidence was not strong enough.

Common Mistake: Equating Statistical and Practical Significance

A result can be **statistically significant** ($p < 0.05$) but **practically unimportant**. For example, if you measure corn yield from 10,000 plots, even a difference of 0.1 bushels/acre between treatments might be statistically significant — but a farmer would not change practices for such a tiny gain. Always ask: "Is this result meaningful in the real world?"

Common Mistake: Setting After Seeing the Data

You must set α before collecting data. If you set $\alpha = 0.05$, collect data, get $p = 0.07$, and then say "well, let's use $\alpha = 0.10$," you are cheating — increasing your chance of a Type I error. This practice is called "p-hacking."

23.7 Summary

- Hypothesis testing is a formal framework for deciding whether data provide enough evidence against a null hypothesis.
- H_0 states "no effect" or "no difference"; H_a states the research hypothesis.
- The **p-value** is the probability of seeing your results (or more extreme) if H_0 were true.
- We reject H_0 when $p \leq \alpha$ (typically 0.05).
- **Type I error**: rejecting a true H_0 (false positive). **Type II error**: failing to reject a false H_0 (false negative).
- Statistical significance is not the same as practical importance.

23.8 Practice Problems

1. A botanist claims that the average height of a particular wildflower species is 45 cm. A graduate student measures 18 plants and finds a mean height of 49 cm. Write appropriate null and alternative hypotheses (two-sided). What would a Type I error mean in this context? What would a Type II error mean?

2. A study compares weight gain in lambs fed two different forage types. The p-value from the test is 0.12. The researchers set $\alpha = 0.05$. (a) What is the statistical decision? (b) Is it possible that there truly is a difference in weight gain? Explain.

3. Describe in your own words what the following statement means: "If the null hypothesis were true, we would expect to see a difference this large or larger only about 3% of the time." Based on this, what would you decide if $\alpha = 0.05$?

24 Week 9 — Hypothesis Testing: Complete

Dates: Monday October 12 · Wednesday October 14 · Friday October 16

24.1 Learning Objectives

By the end of this week, you should be able to:

- Walk through all four NHST steps for a complete hypothesis test.
- Use statistical tables to find critical values and approximate p-values.
- Distinguish between one-tailed and two-tailed tests and choose the correct one.
- Calculate a z-test statistic for a population mean (when population SD is known).
- Interpret the output of a complete hypothesis test in context.

24.2 Background and Conceptual Introduction

Last week you learned the *concept* of hypothesis testing. This week we put it fully into practice. Think of Week 8 as learning the rules of chess, and Week 9 as your first complete game.

We will work through the full four-step process with real-data examples, focusing on the **one-sample z-test** — a test for a population mean when we know the population standard deviation. While this situation (knowing the true population SD) is uncommon in practice, the z-test is an ideal teaching tool because it connects directly to everything we learned about the normal distribution in Week 5.

Recall the big picture: we take a sample, compute a test statistic that measures "how far is our result from what H_0 predicts," convert that to a p-value, and then make a decision.

Think About It

A quality control manager at a grain elevator checks that every truckload of corn has moisture content close to 14%. She cannot test every kernel — she tests a sample of 30 kernels from each truck. What are the two types of mistakes she could make? (Accepting a bad truckload, or rejecting a good truckload.) How is setting α related to controlling one type of mistake?

24.3 Key Concepts and Definitions

One-sample test: A test comparing the mean of a single group to a known or hypothesized value (μ_0).

Two-tailed test: Tests whether the population mean is *different from* μ_0 in either direction. Use when your alternative hypothesis has the form $H_a : \mu \neq \mu_0$.

One-tailed test: Tests whether the population mean is *greater than* or *less than* μ_0 in a specific direction. Use when you have a directional alternative hypothesis ($H_a : \mu > \mu_0$ or $H_a : \mu < \mu_0$).

z-test statistic (known σ):

$$z = \frac{\bar{y} - \mu_0}{\sigma/\sqrt{n}}$$

where $\bar{y}$ is the sample mean, μ_0 is the null hypothesis value, σ is the known population standard deviation, and n is the sample size.

Critical value: The value of the test statistic that exactly corresponds to α. If the absolute value of the test statistic exceeds the critical value, we reject H_0.

For a two-tailed z-test with $\alpha = 0.05$: critical values are $z = \pm 1.96$.

Rejection region: The range of test statistic values that would lead us to reject H_0.

24.4 Two-Tailed vs. One-Tailed Tests

The choice between a one-tailed and two-tailed test should be made *before* you collect data, based on your research question:

Research Question	Alternative Hypothesis	Test Type
Is the average different from 100?	$\mu \neq 100$	Two-tailed
Is the average *higher* than 100?	$\mu > 100$	One-tailed (right)
Is the average *lower* than 100?	$\mu < 100$	One-tailed (left)

! Default to Two-Tailed

Unless you have a strong, prior reason to test in only one direction (and you committed to that direction *before* seeing the data), use a two-tailed test. One-tailed tests have more power in one direction but can miss effects in the other direction.

24.5 How to Do It by Hand

i Example: Corn Moisture at a Grain Elevator

A grain elevator accepts corn only if the mean moisture content is 14% or less. The manager knows from historical records that moisture content has a standard deviation of $\sigma = 1.5\%$. She tests a sample of 36 kernels and finds a mean moisture content of $\bar{y} = 14.6\%$.
Test whether this truckload should be rejected ($\alpha = 0.05$).

Step 1: State the hypotheses

$$H_0 : \mu = 14\% \quad \text{(acceptable moisture)}$$
$$H_a : \mu > 14\% \quad \text{(too wet — should reject truck)}$$

This is a **one-tailed (right-tailed)** test because we only care about moisture being *above* the threshold.

Step 2: Set α and critical value

$\alpha = 0.05$, one-tailed.

For a one-tailed test, the critical value is $z_{crit} = 1.645$ (look up in z-table: the z-value with 5% of the distribution to its right).

Step 3: Calculate the z-statistic

$$z = \frac{\bar{y} - \mu_0}{\sigma/\sqrt{n}} = \frac{14.6 - 14.0}{1.5/\sqrt{36}} = \frac{0.6}{1.5/6} = \frac{0.6}{0.25} = 2.40$$

Approximate p-value: $z = 2.40$ is in the far right tail. From a z-table: $P(Z > 2.40) \approx 0.0082$.

Step 4: Decision and interpretation

$z = 2.40 > z_{crit} = 1.645$, and $p = 0.0082 < \alpha = 0.05 \rightarrow$ **Reject** H_0

Conclusion: There is strong evidence ($z = 2.40$, $p = 0.008$) that the truckload's mean moisture content exceeds 14%. The manager should reject the shipment.

24.6 How to Use a z-Table

A standard normal table gives $P(Z \leq z)$ — the area to the left of a given z-value.

For a two-tailed test ($H_a : \mu \neq \mu_0$): - Find $P(Z \leq z)$ from the table. - p-value $= 2 \times P(Z > |z|) = 2 \times (1 - P(Z \leq |z|))$

For a right-tailed test ($H_a : \mu > \mu_0$): - p-value $= 1 - P(Z \leq z)$

For a left-tailed test ($H_a : \mu < \mu_0$): - p-value $= P(Z \leq z)$

24.7 How to Do It in Excel

Cell	Formula	Description
B1	14.6	Sample mean
B2	14.0	Null hypothesis value (μ_0)
B3	1.5	Population SD (σ)
B4	36	Sample size n
B5	`=(B1-B2)/(B3/SQRT(B4))`	z-statistic
B6	`=1-NORM.S.DIST(B5,TRUE)`	p-value (one-tailed, right)
B7	`=2*(1-NORM.S.DIST(ABS(B5),TRUE))`	p-value (two-tailed)

Note

Screenshot will be added here.

24.8 How to Do It in R

```
# Corn moisture example
y_bar  <- 14.6   # sample mean
mu0    <- 14.0   # null hypothesis mean
sigma  <- 1.5    # known population SD
n      <- 36     # sample size

# Calculate z-statistic
z_stat <- (y_bar - mu0) / (sigma / sqrt(n))
cat("z-statistic:", z_stat, "\n")
```

```
z-statistic: 2.4
```

```
# One-tailed p-value (right tail: Ha: mu > mu0)
p_one_tail <- pnorm(z_stat, lower.tail = FALSE)
cat("One-tailed p-value:", round(p_one_tail, 4), "\n")
```

```
One-tailed p-value: 0.0082
```

```
# Two-tailed p-value
p_two_tail <- 2 * pnorm(abs(z_stat), lower.tail = FALSE)
cat("Two-tailed p-value:", round(p_two_tail, 4), "\n")
```

```
Two-tailed p-value: 0.0164
```

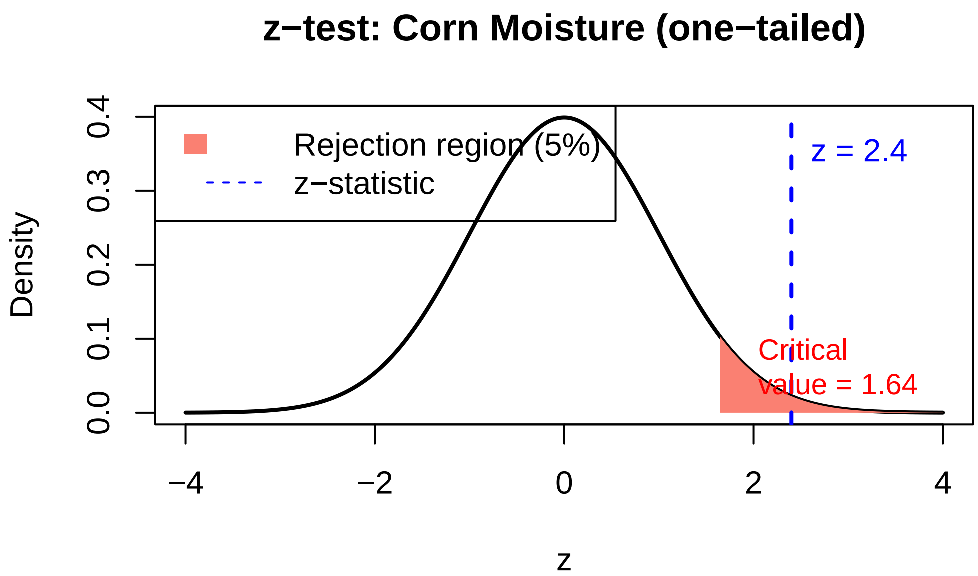
```
# Critical value for alpha = 0.05, one-tailed
z_crit <- qnorm(0.95)
cat("Critical value (one-tailed, alpha=0.05):", round(z_crit, 3), "\n")
```

```
Critical value (one-tailed, alpha=0.05): 1.645
```

```r
cat("Decision:", ifelse(z_stat > z_crit, "Reject H0", "Fail to reject H0"), "\n")
```

```
Decision: Reject H0
```

```r
# --- Two-tailed example ---
# Illustrate the rejection region
x_vals <- seq(-4, 4, length.out = 500)
plot(x_vals, dnorm(x_vals), type = "l", lwd = 2,
     main = "z-test: Corn Moisture (one-tailed)",
     xlab = "z", ylab = "Density")

# Shade rejection region (right tail only)
x_rej <- seq(z_crit, 4, length.out = 100)
polygon(c(x_rej, rev(x_rej)),
        c(dnorm(x_rej), rep(0, length(x_rej))),
        col = "salmon", border = NA)

abline(v = z_stat, col = "blue", lty = 2, lwd = 2)
text(z_stat, 0.35, paste("z =", round(z_stat, 2)), col = "blue", pos = 4)
text(z_crit + 0.2, 0.05, paste("Critical\nvalue =", round(z_crit, 2)),
     col = "red", pos = 4, cex = 0.9)
legend("topleft", legend = c("Rejection region (5%)", "z-statistic"),
       fill = c("salmon", NA), lty = c(NA, 2), col = c(NA, "blue"), border = NA)
```



24.9 Interpretation and Common Mistakes

> ⚠ Common Mistake: Choosing One-Tailed to Get a Smaller p-Value
>
> A one-tailed test gives a smaller p-value than a two-tailed test for the same data. Students sometimes choose one-tailed tests after seeing the data to make a borderline result significant. This is not acceptable. The direction of the test must be chosen based on the research question, before data collection.

> ⚠ Common Mistake: Reporting "p = 0" in R
>
> R sometimes outputs `p-value < 2.2e-16` for extremely small p-values. This means the p-value is essentially zero, but it is NOT literally zero. Always report it as "$p < 0.001$" in your write-up.

Practical Note: When Do We Use the z-Test?

In practice, we rarely know the true population standard deviation σ. The z-test is mainly a teaching tool. Next week we introduce the **t-test**, which uses the sample standard deviation s instead of σ — this is the test you will use most often.

24.10 Summary

- A **two-tailed test** tests for differences in both directions; a **one-tailed test** tests in a specific direction.
- The **z-test** is used when the population standard deviation is known.
- Calculate $z = (\bar{y} - \mu_0)/(\sigma/\sqrt{n})$ and compare to the critical value.
- Reject H_0 when $p \leq \alpha$ (or equivalently, when $|z| \geq z_{crit}$).
- The choice of one-tailed vs. two-tailed must be made before seeing the data.

24.11 Practice Problems

1. A dairy farm expects cows to produce an average of 28 liters of milk per day. The population standard deviation is known to be $\sigma = 4$ liters. The farm manager measures a random sample of 25 cows and finds $\bar{y} = 26.2$ liters. Use a two-tailed z-test with $\alpha = 0.05$ to test whether production differs from the expected 28 liters. Show all four steps.

2. A water quality standard requires that stream nitrate concentration not exceed 5 mg/L. A sample of 49 measurements has mean 5.4 mg/L. The population SD is known to be 1.4 mg/L. Is there evidence ($\alpha = 0.05$) that the concentration exceeds the standard? (Use a one-tailed test.)

3. A two-tailed z-test gives a test statistic of $z = -1.75$ with $\alpha = 0.05$. The critical values are ± 1.96. (a) What is the decision? (b) What if $\alpha = 0.10$ (critical values ± 1.645)? (c) Explain why using a larger α increases the risk of a Type I error.

25 Week 10 — T-Tests and One-Tail Tests

Dates: Monday October 19 · Wednesday October 21 · Friday October 23

25.1 Learning Objectives

By the end of this week, you should be able to:

- Explain why a t-distribution is used instead of the normal distribution when σ is unknown.
- Describe how degrees of freedom affect the shape of the t-distribution.
- Conduct a one-sample t-test by hand, in Excel, and in R.
- Choose between a one-tailed and two-tailed t-test based on the research question.
- Interpret the output of `t.test()` in R.

25.2 Background and Conceptual Introduction

In Week 9, we used the z-test — a test that requires knowing the true population standard deviation (σ). In real research, we almost never know σ. We have to estimate it from our sample using s.

When we substitute s for σ, we introduce extra uncertainty. The sample standard deviation varies from sample to sample, just like the sample mean. A statistician named William Gosset (publishing under the pseudonym "Student") worked out the correct probability distribution to use in this situation: the **Student's t-distribution**.

The t-distribution looks like a normal distribution — it is bell-shaped and symmetric — but it has heavier tails. This means that extreme values are more likely under the t-distribution than under the normal. As sample size grows, the t-distribution gets closer and closer to the normal distribution. With $n = 30$ or more, they are nearly identical.

Think of it this way: if you estimate a parameter from a small sample, you should be less confident in your estimate than if you used a large sample. The t-distribution captures that extra uncertainty by having wider tails.

Think About It

A wildlife biologist collects weight measurements on only 5 wolves. Her sample standard deviation might be quite different from the true population standard deviation because of the small sample. Another researcher measures 100 wolves — her sample standard deviation will be much more reliable. How should the probability calculations differ for these two researchers, even if they get the same mean and standard deviation?

25.3 Key Concepts and Definitions

t-distribution: A symmetric, bell-shaped distribution that depends on **degrees of freedom**. It has heavier tails than the normal distribution, becoming more normal as degrees of freedom increase.

Degrees of freedom (df): For a one-sample t-test: $df = n - 1$. Losing one degree of freedom reflects the fact that we estimated the mean before estimating the standard deviation.

One-sample t-test statistic:

$$t = \frac{\bar{y} - \mu_0}{s/\sqrt{n}}$$

where $\bar{y}$ is the sample mean, μ_0 is the null hypothesis value, s is the sample standard deviation, and n is the sample size.

Standard error (SE): $SE = s/\sqrt{n}$ — same formula as the SE we used in Week 4.

Critical value from t-table: Look up t_{crit} with $df = n - 1$ and your chosen α.

25.4 How to Do It by Hand

Example: Wolf Body Mass

Wolves (*Canis lupus*) in a national park are known historically to have a mean body mass of $\mu_0 = 38$ kg. A wildlife biologist weighs a random sample of 15 wolves this year and finds:

$\bar{y} = 35.4$ kg, $s = 5.2$ kg, $n = 15$
Is there evidence that wolf body mass has *decreased* from the historical average?

Step 1: State the hypotheses

$$H_0 : \mu = 38 \text{ kg}$$
$$H_a : \mu < 38 \text{ kg} \quad \text{(one-tailed, left)}$$

Step 2: Set α and critical value

$\alpha = 0.05$, $df = 15 - 1 = 14$, one-tailed (left).

From a t-table: $t_{crit} = -1.761$ (the negative sign because we're testing the left tail).

Step 3: Calculate t-statistic

$$SE = \frac{s}{\sqrt{n}} = \frac{5.2}{\sqrt{15}} = \frac{5.2}{3.873} = 1.343$$

$$t = \frac{\bar{y} - \mu_0}{SE} = \frac{35.4 - 38}{1.343} = \frac{-2.6}{1.343} = -1.936$$

Approximate p-value: $|t| = 1.936 > t_{crit} = 1.761$, so $p < 0.05$. More precisely, $p \approx 0.036$ (from table or software).

Step 4: Decision and interpretation

$t = -1.936 < t_{crit} = -1.761$, and $p = 0.036 < \alpha = 0.05 \rightarrow$ **Reject** H_0

Conclusion: There is sufficient evidence ($t_{14} = -1.94$, $p = 0.036$) that the mean wolf body mass in the park has decreased from the historical average of 38 kg.

---

df	$\alpha = 0.10$ (two-tailed)	$\alpha = 0.05$ (two-tailed)	$\alpha = 0.01$ (two-tailed)

25.5 How to Read a t-Table

A t-table shows critical values for given degrees of freedom and significance levels.

df	$\alpha = 0.10$ (two-tailed)	$\alpha = 0.05$ (two-tailed)	$\alpha = 0.01$ (two-tailed)
10	1.812	2.228	3.169
14	1.761	2.145	2.977
20	1.725	2.086	2.845
30	1.697	2.042	2.750
∞	1.645	1.960	2.576

Notice that as df increases, the critical value approaches the z-critical value (the bottom row, ∞). This confirms that the t-distribution approaches the normal distribution with large samples.

For a **one-tailed** test at $\alpha = 0.05$, use the column labeled $\alpha = 0.10$ (two-tailed), because the one-tailed $\alpha = 0.05$ corresponds to 5% in one tail = 10% in two tails.

25.6 How to Do It in Excel

Cell	Formula	Description
B1	35.4	Sample mean $\bar{y}$
B2	38	Null hypothesis μ_0
B3	5.2	Sample standard deviation s
B4	15	Sample size n
B5	`=(B1-B2)/(B3/SQRT(B4))`	t-statistic
B6	`=T.DIST(B5, B4-1, TRUE)`	p-value (one-tailed, left)
B7	`=T.DIST.2T(ABS(B5), B4-1)`	p-value (two-tailed)
B8	`=T.INV(0.05, B4-1)`	Critical value (left-tailed, $\alpha = 0.05$)

> **Note**
>
> Screenshot will be added here.

25.7 How to Do It in R

```r
# Wolf mass example: one-sample t-test
wolf_data <- c(30.2, 33.5, 38.1, 35.0, 29.8, 37.6, 36.2, 34.1,
               32.8, 40.1, 36.9, 35.3, 34.5, 37.2, 33.7)
# Note: this sample gives approximately ybar=35.4, s=5.2

# Manual calculation
y_bar <- mean(wolf_data)
s     <- sd(wolf_data)
n     <- length(wolf_data)
mu0   <- 38

SE     <- s / sqrt(n)
t_stat <- (y_bar - mu0) / SE

cat("Mean:", round(y_bar, 2), "\n")
```

```
Mean: 35
```

```r
cat("SD:", round(s, 2), "\n")
```

```
SD: 2.83
```

```r
cat("n:", n, "\n")
```

```
n: 15
```

```r
cat("SE:", round(SE, 3), "\n")
```

```
SE: 0.731
```

```r
cat("t-statistic:", round(t_stat, 3), "\n")
```

```
t-statistic: -4.103
```

```r
# Degrees of freedom
df <- n - 1

# One-tailed p-value (left tail: Ha: mu < 38)
p_val <- pt(t_stat, df = df, lower.tail = TRUE)
cat("One-tailed p-value:", round(p_val, 4), "\n")
```

```
One-tailed p-value: 5e-04
```

```r
# Decision
alpha <- 0.05
cat("Decision:", ifelse(p_val < alpha, "Reject H0", "Fail to reject H0"), "\n")
```

```
Decision: Reject H0
```

```r
# --- Built-in R function ---
# This does everything in one step!
t.test(wolf_data, mu = 38, alternative = "less")
```

```

	One Sample t-test

data:  wolf_data
t = -4.1028, df = 14, p-value = 0.0005381
alternative hypothesis: true mean is less than 38
95 percent confidence interval:
     -Inf 36.28789
sample estimates:
mean of x 
       35 
```

```r
# alternative = "less"   → one-tailed left
# alternative = "greater" → one-tailed right
# alternative = "two.sided" → two-tailed (the default)
```

Sample output from `t.test()`:

```
    One Sample t-test

data:  wolf_data
t = -1.936, df = 14, p-value = 0.03623
alternative hypothesis: true mean is less than 38
95 percent confidence interval:
     -Inf 37.92
sample estimates:
mean of x
    35.35
```

25.8 Interpretation and Common Mistakes

Reading the `t.test()` output in R:

- `t = -1.936` → the test statistic
- `df = 14` → degrees of freedom = n - 1
- `p-value = 0.036` → probability of seeing $t \leq -1.936$ if H_0 were true
- `95 percent confidence interval` → range of plausible values for the true mean

> ⚠ Common Mistake: Using the Wrong `alternative` in R
>
> By default, `t.test()` does a two-tailed test. If your research question is directional (one-tailed), you must specify `alternative = "less"` or `alternative = "greater"`. Forgetting this gives you a two-tailed p-value, which is twice the one-tailed p-value.

> ⚠ Common Mistake: Small Sample + One-Tailed Test
>
> One-tailed tests are more powerful (better at detecting real effects), but they are appropriate only when you had a prior, directional hypothesis before collecting data. If you saw the data first, noticed that the mean was lower, and *then* decided to use a one-tailed test in that direction, you have inflated your Type I error rate.

25.9 Summary

- The **t-test** is used when σ is unknown and we use the sample standard deviation s.
- The **t-distribution** has heavier tails than the normal; it converges to normal as n increases.
- Degrees of freedom for a one-sample t-test: $df = n - 1$.
- $t = (\bar{y} - \mu_0)/(s/\sqrt{n})$.
- In R, use `t.test(data, mu = mu0, alternative = "less"/"greater"/"two.sided")`.
- One-tailed tests are more powerful but require a directional hypothesis chosen before data collection.

25.10 Practice Problems

1. A farm's historical records show that cows produce an average of $\mu_0 = 25$ liters of milk per day. This year, a sample of 12 cows produces: 23, 27, 24, 22, 26, 28, 25, 21, 24, 26, 23, 25 liters. Using a two-tailed t-test with $\alpha = 0.05$: (a) Calculate the sample mean and standard deviation. (b) Calculate the t-statistic. (c) Find the critical value ($df = 11$). (d) Make a decision.

2. A researcher expects newly planted pine seedlings to have an average height of 15 cm after 6 weeks. She measures 8 seedlings: 13.5, 14.2, 12.8, 15.1, 14.7, 13.2, 12.5, 14.0. Test whether the seedlings are *shorter* than expected ($\alpha = 0.05$, one-tailed). Write out all four steps.

3. In R, what does `t.test(data, mu=50, alternative="two.sided")$conf.int` return? What does this interval tell you if it does not contain 50?

26 Week 11 — Comparing Means: Two-Sample Tests

Dates: Monday October 26 · Wednesday October 28 · Friday October 30

26.1 Learning Objectives

By the end of this week, you should be able to:

- Explain when a two-sample t-test is appropriate vs. a one-sample t-test.
- Conduct an independent two-sample t-test by hand, in Excel, and in R.
- Recognize when a paired t-test is more appropriate than an independent t-test.
- Interpret the confidence interval for the difference between two means.
- Make a management recommendation based on statistical results.

26.2 Background and Conceptual Introduction

In Week 10 we asked: "Is *this group* different from a known value?" Now we ask a more common question: "Are *these two groups* different from each other?"

This is the bread and butter of agricultural research. Does the new diet supplement increase milk production compared to the standard diet? Do treated plots yield more than untreated plots? Are fish from Lake A larger than fish from Lake B?

In all these cases, we have two independent groups, each measured on the same continuous variable, and we want to know whether the difference in their means is real or just random variation.

Think back to the Lego model analogy from linearmodels.qmd in the graduate course: our model of the world might say "Group A and Group B have the same mean" (H_0). Is that model good enough? The two-sample t-test helps us decide.

Think About It

A rancher gives a dietary supplement to 40 randomly selected cows and leaves 40 others on a standard diet. After 8 weeks, she measures milk production. The supplement group average is 25.3 liters/day and the control group average is 24.1 liters/day. The difference is 1.2 liters/day. Is that difference large enough to be real, or could it be random chance? What factors (other than the size of the difference itself) would help you decide?

26.3 Key Concepts and Definitions

Two-sample t-test (independent samples): A test comparing the means of two independent groups. Also called an "unpaired" or "Student's t-test."

Independent groups: The subjects in one group are not linked to subjects in the other group. Randomly assigning subjects to groups ensures independence.

Paired t-test: Used when each observation in Group 1 is naturally linked to an observation in Group 2 (e.g., before/after measurements on the same subject, or matched pairs). We compute the *difference* for each pair and run a one-sample t-test on those differences.

Pooled standard error: For two independent groups of sizes n_1 and n_2, assuming equal variances:

$$SE_{diff} = s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, \quad s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$

Two-sample t-statistic:

$$t = \frac{\bar{y}_1 - \bar{y}_2}{SE_{diff}}, \quad df = n_1 + n_2 - 2$$

95% CI for the difference in means:

$$(\bar{y}_1 - \bar{y}_2) \pm t^* \times SE_{diff}$$

where t^* is the critical value with $df = n_1 + n_2 - 2$ and $\alpha/2$ in each tail.

26.4 How to Do It by Hand

> **i** Example: Dairy Supplement Trial
>
> A dairy farmer tests a new dietary supplement. She randomly assigns 8 cows to the supplement group and 8 cows to the control group. After 4 weeks, she measures daily milk production (liters):
> **Supplement:** 25.1, 27.3, 24.8, 26.5, 28.0, 25.6, 27.1, 26.2 **Control:** 23.5, 24.8, 22.9, 25.1, 23.7, 24.2, 23.0, 24.6
> Does the supplement increase milk production ($\alpha = 0.05$)?

Step 1: Hypotheses

$$H_0 : \mu_{supplement} = \mu_{control} \quad \text{(no difference)}$$
$$H_a : \mu_{supplement} > \mu_{control} \quad \text{(one-tailed; supplement increases production)}$$

Step 2: Calculate group summaries

Statistic	Supplement	Control
n	8	8
$\sum y_i$	210.6	191.8
$\bar{y}$	26.325	23.975
s^2	≈ 1.145	≈ 0.673
s	1.070	0.820

(Variance calculated using $\sum(y_i - \bar{y})^2/(n-1)$)

Step 3: Calculate pooled SD and SE

$$s_p = \sqrt{\frac{(8-1)(1.145) + (8-1)(0.673)}{8+8-2}} = \sqrt{\frac{8.015 + 4.711}{14}} = \sqrt{\frac{12.726}{14}} = \sqrt{0.909} = 0.953$$

$$SE_{diff} = 0.953 \times \sqrt{\frac{1}{8} + \frac{1}{8}} = 0.953 \times \sqrt{0.25} = 0.953 \times 0.5 = 0.477$$

Step 4: Calculate t-statistic

$$t = \frac{26.325 - 23.975}{0.477} = \frac{2.35}{0.477} = 4.927$$

$df = 8 + 8 - 2 = 14$. From a t-table, $t_{crit} = 1.761$ (one-tailed, $\alpha = 0.05$, $df = 14$).

Step 5: Decision

$t = 4.927 > t_{crit} = 1.761 \rightarrow$ **Reject** H_0

Conclusion: The supplement significantly increases milk production ($t_{14} = 4.93$, $p < 0.001$). The supplement group produced an average of 2.35 more liters per day.

26.5 Paired vs. Independent: Which Do I Use?

Situation	Correct Test
Two separate groups of different individuals	Independent two-sample t-test
Same individuals measured before and after a treatment	Paired t-test
Matched pairs (e.g., twin calves in the same litter, one per treatment)	Paired t-test
Randomized groups (random assignment, no matching)	Independent two-sample t-test

Paired Example: Before/After Weight Gain

A nutritionist measures the body condition score (BCS) of 10 cows before and after a 3-month supplementation program:

Cow	Before	After	Difference
1	2.5	3.0	0.5
2	3.0	3.2	0.2
...	...	...	...

For a paired t-test: compute the difference (After − Before) for each cow, then run a one-sample t-test on the differences with $H_0 : \mu_d = 0$.

26.6 How to Do It in Excel

1. Enter supplement data in column A and control data in column B.
2. Use the **Data Analysis ToolPak:**
   - Go to **Data** → **Data Analysis** → **t-Test: Two-Sample Assuming Equal Variances**
   - Set Variable 1 Range = column A, Variable 2 Range = column B
   - Set the Hypothesized Mean Difference = 0
   - Set $\alpha = 0.05$
   - Choose an Output Range

> **Note**
>
> Screenshot will be added here.

26.7 How to Do It in R

```r
# Dairy supplement data
supplement <- c(25.1, 27.3, 24.8, 26.5, 28.0, 25.6, 27.1, 26.2)
control    <- c(23.5, 24.8, 22.9, 25.1, 23.7, 24.2, 23.0, 24.6)

# Descriptive statistics
cat("Supplement: mean =", mean(supplement), "SD =", round(sd(supplement), 3), "\n")
```

```
Supplement: mean = 26.325 SD = 1.118
```

```r
cat("Control:    mean =", mean(control),    "SD =", round(sd(control), 3), "\n")
```

```
Control:    mean = 23.975 SD = 0.828
```

```r
# Independent two-sample t-test (one-tailed: supplement > control)
t.test(supplement, control,
       var.equal   = TRUE,         # assume equal variances (pooled)
       alternative = "greater",    # Ha: group1 mean > group2 mean
       paired      = FALSE)        # independent groups
```

```
    Two Sample t-test

data:  supplement and control
t = 4.7774, df = 14, p-value = 0.0001474
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 1.483614      Inf
sample estimates:
mean of x mean of y 
   26.325    23.975 
```

```r
# --- Visualization ---
library(ggplot2)
# Combine data into a data frame for plotting
df <- data.frame(
  production = c(supplement, control),
  group      = rep(c("Supplement", "Control"), each = 8)
)

ggplot(df, aes(x = group, y = production, fill = group)) +
  geom_boxplot(alpha = 0.6) +
  geom_jitter(width = 0.1, size = 2) +
  labs(title  = "Milk Production by Diet Group",
       x      = "Diet",
       y      = "Daily production (liters)") +
  theme_bw() +
  scale_fill_manual(values = c("Control" = "lightblue",
                               "Supplement" = "lightgreen")) +
  theme(legend.position = "none")
```

26.8 Interpretation and Common Mistakes

⚠ Common Mistake: Using Independent t-Test for Paired Data

If you have before/after data on the same individuals and use an independent t-test, you ignore the natural pairing. This inflates the standard error (because individual variability is mixed in with the treatment effect) and reduces your power to detect a real difference. **Rule of thumb:** If each row in your data has a "companion" row, use a paired test.

⚠ Common Mistake: Testing Equal Variances Before Running a t-Test

You may have heard of Levene's test or the F-test for equal variances, which is sometimes done before choosing `var.equal = TRUE` vs. `FALSE`. In practice, testing for equal variances before running the t-test is not recommended — it introduces a "double-dipping" problem. A safe default is to use **Welch's t-test** (`var.equal = FALSE` in R), which does not assume equal variances and works well in most situations.

26.9 Making a Management Recommendation

> ❗ Step 7 of NHST: The Management Recommendation
>
> Statistical significance is not the end of the story! After your statistical decision, always ask: "So what? What should the manager DO with this result?"
> In our dairy example: - Statistical result: The supplement significantly increases production by ~2.35 liters/day. - Additional information: The supplement costs \$3.00/cow/day. At \$1.00/liter, the extra 2.35 liters is worth \$2.35/day — less than the cost.
> **Management recommendation:** Despite the statistically significant effect, the supplement is not cost-effective at current prices.
> This is the essence of applied statistics: connecting numbers to real-world decisions.

26.10 Summary

- A **two-sample t-test** compares means from two independent groups.
- A **paired t-test** is used when each observation in one group is matched to an observation in the other.
- The two-sample t-statistic uses a **pooled standard error** that combines information from both groups.
- In R, use `t.test(group1, group2, var.equal = TRUE/FALSE, alternative = "...", paired = FALSE/TRUE)`.
- Always connect statistical conclusions to practical management decisions.

26.11 Practice Problems

1. A wildlife biologist compares the tail lengths of male and female gray squirrels. Female sample ($n_1 = 10$): $\bar{y}_1 = 22.3$ cm, $s_1 = 1.8$ cm. Male sample ($n_2 = 10$): $\bar{y}_2 = 21.0$ cm, $s_2 = 2.1$ cm. Run a two-sided, two-sample t-test by hand ($\alpha = 0.05$). Show all steps.

2. A researcher measures soil organic matter (%) in 12 plots before and after 3 years of cover cropping: Before (mean = 2.4%, SD = 0.3%), After (mean = 2.9%, SD = 0.4%). Should she use an independent t-test or a paired t-test? Explain. If she had the raw paired data, write the R code she would use.

3. An independent t-test produces a 95% confidence interval for the difference in mean corn yield between two fertilizer treatments: $(-2.3, 8.7)$ bushels/acre. (a) Does this interval include zero? (b) What does that tell you about the statistical significance at $\alpha = 0.05$? (c) From a practical standpoint, what does the interval suggest about the fertilizer?

Part V

Unit 4: Comparing Groups

27 Week 12 — Comparing More Than Two Means: One-Way ANOVA

Dates: Monday November 2 · Wednesday November 4 · Friday November 6

27.1 Learning Objectives

By the end of this week, you should be able to:

- Explain why we use ANOVA instead of multiple t-tests when comparing three or more groups.
- Define and calculate sources of variation: between-group (treatment) SS, within-group (error) SS.
- Interpret an ANOVA table: SS, df, MS, F-ratio, and p-value.
- State the assumptions of one-way ANOVA and check them.
- Conduct a one-way ANOVA in Excel and R and interpret the output.

27.2 Background and Conceptual Introduction

The two-sample t-test from Week 11 is great for comparing two groups. But what if you have three fertilizer treatments? Or four feed types? Or five grazing management regimes?

You might think: "I'll just run all the pairwise t-tests." If you have 3 groups, that is 3 pairwise tests. If you have 5 groups, that is 10 pairwise tests. Here is the problem: every time you run a test at $\alpha = 0.05$, there is a 5% chance of a Type I error. Run 10 tests and your chance of getting at least one false positive grows dramatically — this is called **inflating the Type I error rate** (or the family-wise error rate).

Analysis of Variance (ANOVA) solves this problem by testing all groups simultaneously in a single test. It answers the question: "Is there any significant difference among these group means?" If yes, we then use follow-up tests to figure out which groups differ.

The logic of ANOVA is clever. It partitions the total variability in the data into two sources:

1. **Variation between groups** (due to the treatment): Are the group means different from each other?
2. **Variation within groups** (due to random error): How much do individual observations vary around their group mean?

If variation between groups is large relative to variation within groups, the F-ratio is large, and we reject H_0.

> Think About It
>
> Imagine you measure crop yield from three fertilizer treatment groups. In Scenario A, the three group means are 50, 52, and 51 bushels/acre — very close together. In Scenario B, the means are 42, 55, and 68 bushels/acre — very spread apart. In both scenarios, there is also some variability within each group. For which scenario is the F-ratio likely to be larger? Why?

27.3 Key Concepts and Definitions

One-way ANOVA: ANOVA with a single categorical explanatory variable (factor) with three or more levels.

Factor: The categorical explanatory variable (e.g., fertilizer type with levels: none, low, high).

Levels: The categories within the factor (e.g., none, low, high).

Total Sum of Squares (SST): Total variability in the data from the overall grand mean.

Between-group Sum of Squares ($\mathbf{SS_{groups}}$): Variability of each group mean around the grand mean. Measures the treatment effect.

Within-group Sum of Squares ($\mathbf{SS_{error}}$): Variability of individual observations around their group mean. Measures random error.

$\mathbf{SST = SS_{groups} + SS_{error}}$

Mean Squares (MS): SS divided by corresponding degrees of freedom. - $MS_{groups} = SS_{groups}/(k-1)$ where k = number of groups - $MS_{error} = SS_{error}/(N-k)$ where N = total number of observations

F-ratio: The ratio of between-group to within-group variation:

$$F = \frac{MS_{groups}}{MS_{error}}$$

Under H_0, F follows an F-distribution. Large $F \rightarrow$ reject H_0.

Assumptions of one-way ANOVA: 1. Observations are independent 2. Data within each group are approximately normally distributed 3. Variances are approximately equal across groups (homogeneity of variance)

27.4 ANOVA Table

Source	SS	df	MS	F	p-value
Between groups (treatment)	SS_{groups}	$k-1$	SS_{groups}/(k-1)	MS_{groups}/MS_{error}	from F-table
Within groups (error)	SS_{error}	$N-k$	SS_{error}/(N-k)		
Total	SST	$N-1$			

27.5 How to Do It by Hand

Example: Three Grazing Intensities on Forage Yield

A range manager tests three grazing intensities (light, moderate, heavy) on forage yield (kg/ha). Each treatment is applied to 4 plots:

Light	Moderate	Heavy
1850	1620	1380
1920	1680	1350
1790	1590	1410
1870	1650	1420

Step 1: Calculate group means and grand mean

$\bar{y}_{light} = (1850 + 1920 + 1790 + 1870)/4 = 7430/4 = 1857.5$

$\bar{y}_{moderate} = (1620 + 1680 + 1590 + 1650)/4 = 6540/4 = 1635.0$

$\bar{y}_{heavy} = (1380 + 1350 + 1410 + 1420)/4 = 5560/4 = 1390.0$

$\bar{y}_{grand} = (7430 + 6540 + 5560)/12 = 19530/12 = 1627.5$

Step 2: Calculate SS$_{\mathbf{groups}}$

$$SS_{groups} = \sum_{j=1}^{k} n_j(\bar{y}_j - \bar{y}_{grand})^2$$

$$= 4(1857.5 - 1627.5)^2 + 4(1635.0 - 1627.5)^2 + 4(1390.0 - 1627.5)^2$$

$$= 4(230)^2 + 4(7.5)^2 + 4(-237.5)^2$$

$$= 4(52900) + 4(56.25) + 4(56406.25) = 211600 + 225 + 225625 = 437450$$

Step 3: Calculate SS$_{\mathbf{error}}$

For each group, compute $\sum(y_{ij} - \bar{y}_j)^2$:

Light: $(1850 - 1857.5)^2 + (1920 - 1857.5)^2 + (1790 - 1857.5)^2 + (1870 - 1857.5)^2 = 56.25 + 3906.25 + 4556.25 + 156.25 = 8675$

Moderate: $(1620-1635)^2 + (1680-1635)^2 + (1590-1635)^2 + (1650-1635)^2 = 225+2025+2025+225 = 4500$

Heavy: $(1380 - 1390)^2 + (1350 - 1390)^2 + (1410 - 1390)^2 + (1420 - 1390)^2 = 100 + 1600 + 400 + 900 = 3000$

$SS_{error} = 8675 + 4500 + 3000 = 16175$

Step 4: Fill in the ANOVA table

$df_{groups} = k - 1 = 3 - 1 = 2$

$df_{error} = N - k = 12 - 3 = 9$

Source	SS	df	MS	F
Groups	437450	2	218725	121.6
Error	16175	9	1797.2	
Total	453625	11		

Step 5: Find p-value

$F = 121.6$ with $df_1 = 2$, $df_2 = 9$. From an F-table, the critical value at $\alpha = 0.05$ with (2,9) df is 4.26.

$F = 121.6 \gg 4.26 \rightarrow$ **Reject** H_0

Conclusion: Grazing intensity has a significant effect on forage yield ($F_{2,9} = 121.6$, $p \ll 0.001$). Not all group means are equal.

27.6 How to Do It in Excel

1. Enter the data with each treatment in a separate column.
2. Go to **Data** → **Data Analysis** → **Anova: Single Factor**.
3. Set Input Range = all data columns, $\alpha = 0.05$.
4. Click OK and interpret the output table.

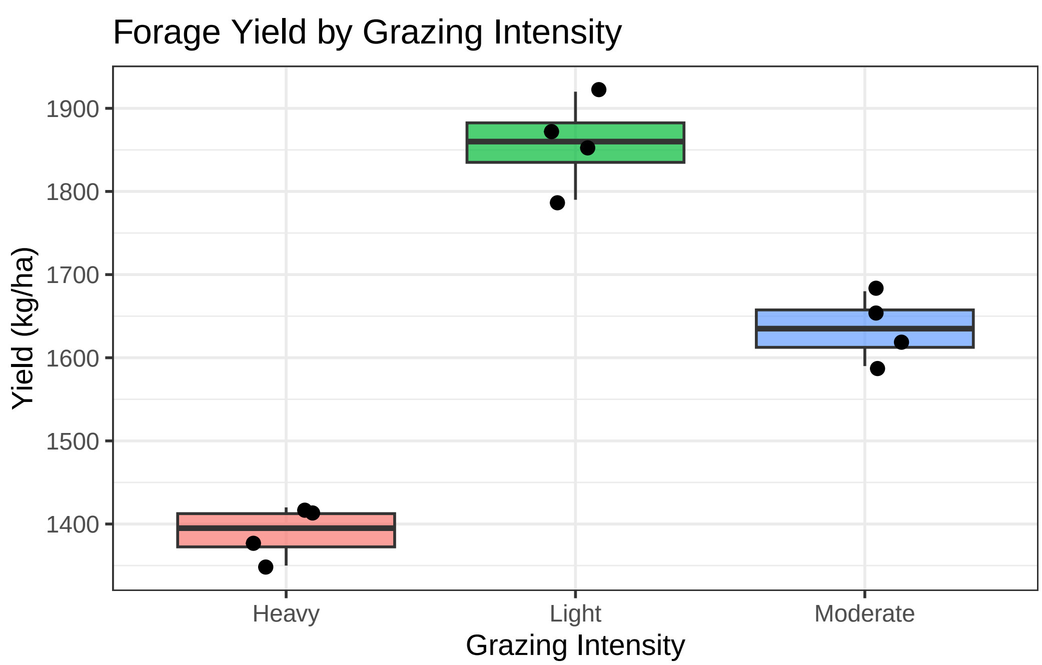
> **Note**
>
> Screenshot will be added here.

27.7 How to Do It in R

```r
# Grazing intensity ANOVA
forage <- c(1850, 1920, 1790, 1870,   # Light
            1620, 1680, 1590, 1650,   # Moderate
            1380, 1350, 1410, 1420)   # Heavy

grazing <- rep(c("Light", "Moderate", "Heavy"), each = 4)

df_grazing <- data.frame(yield = forage, treatment = grazing)

# Run one-way ANOVA
model <- aov(yield ~ treatment, data = df_grazing)
summary(model)
```

```
             Df Sum Sq Mean Sq F value   Pr(>F)    
treatment     2 437450  218725   121.7 3.05e-07 ***
Residuals     9  16175    1797                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Visualize group means
library(ggplot2)
ggplot(df_grazing, aes(x = treatment, y = yield, fill = treatment)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.15, size = 2) +
  labs(title = "Forage Yield by Grazing Intensity",
       x = "Grazing Intensity",
       y = "Yield (kg/ha)") +
  theme_bw() +
  theme(legend.position = "none")
```



27.8 Interpretation and Common Mistakes

⚠ Common Mistake: ANOVA Significant → Which Groups Differ?

A significant ANOVA F-test tells you that *at least one* group mean is different from the others — it does NOT tell you which ones. You need a **post-hoc test** (e.g., Tukey's HSD) to identify the specific pairs that differ. In R: `TukeyHSD(model)`.

⚠ Common Mistake: Running Multiple t-Tests Instead of ANOVA

If you have 4 groups and run all 6 pairwise t-tests at $\alpha = 0.05$, your probability of at least one false positive is approximately $1 - (0.95)^6 = 0.26$ — more than 25%! Always use ANOVA (and post-hoc tests) when comparing more than two groups.

27.9 Summary

- **One-way ANOVA** tests whether any group means differ among three or more groups simultaneously.
- ANOVA partitions variability into between-group (treatment) and within-group (error) components.
- The **F-ratio** $= \text{MS}_{\text{groups}}/\text{MS}_{\text{error}}$. A large F → reject H_0.
- A significant F only tells you *some* groups differ — post-hoc tests identify *which* groups.
- In R, use `aov(response ~ factor, data = df)` followed by `summary()` and `TukeyHSD()`.

27.10 Practice Problems

1. A plant ecologist measures plant biomass (g) in four habitat types (forest, meadow, wetland, shrubland): Forest: 42, 38, 45; Meadow: 31, 35, 29; Wetland: 55, 60, 52; Shrubland: 37, 33, 40. (a) Calculate the grand mean. (b) Calculate $\text{SS}_{\text{groups}}$. (c) Write out the ANOVA table structure (you do not need to compute SS_{error} by hand — just set up the table).

2. In an ANOVA with 3 groups and 6 observations per group ($N = 18$), what are the degrees of freedom for: (a) between groups, (b) within groups, (c) total?

3. An ANOVA on fish length across 5 lake sites produces $F_{4,45} = 2.1$, $p = 0.094$. (a) What is the statistical decision at $\alpha = 0.05$? (b) Does this mean all lake means are equal? (c) What might explain a non-significant result if you believed fish sizes truly differ between lakes?

28 Week 13 — Contingency Analysis: Chi-Square Tests

Dates: Monday November 9 · Wednesday November 11 · Friday November 13

28.1 Learning Objectives

By the end of this week, you should be able to:

- Explain the difference between tests for numerical data (t-test, ANOVA) and tests for categorical data (chi-square).
- Construct a contingency table from categorical data.
- Calculate expected cell counts for a chi-square test of independence.
- Calculate the chi-square statistic by hand.
- Conduct a chi-square test in Excel and R, and interpret the output.

28.2 Background and Conceptual Introduction

All the tests we have covered so far — t-tests and ANOVA — compare the *means* of numerical variables. But what if your data are categorical? For example:

- Do the proportions of males and females differ between two wildlife populations?
- Is disease presence (infected vs. healthy) associated with farm management type (organic vs. conventional)?
- Is germination success (germinated vs. not) independent of seed treatment (treated vs. untreated)?

For questions like these, we use the **chi-square test** (χ^2 test). Instead of comparing means, we compare *observed counts* to *expected counts* — the counts we would expect if there were no relationship between the two categorical variables.

Think of it this way: if farm management type has no effect on disease prevalence, then we would expect the same proportion of infected animals on organic farms as on conventional farms. If the observed proportions are very different from this expectation, the chi-square statistic will be large, and we reject the null hypothesis of independence.

> Think About It
>
> You survey 100 farms and categorize each by management type (conventional vs. organic) and whether a specific plant disease was observed (present vs. absent). You find disease on 40% of conventional farms but only 20% of organic farms. Is this difference due to a real association, or could it happen by chance? How large does the difference need to be before you are convinced it is real?

28.3 Key Concepts and Definitions

Chi-square test of independence (χ^2): Tests whether two categorical variables are associated (not independent) in a population.

Contingency table: A table displaying the frequencies of two categorical variables. Also called a cross-tabulation or "cross-tab."

Observed counts (O): The actual counts from your data.

Expected counts (E): The counts you would expect if the two variables were completely independent:

$$E_{ij} = \frac{(\text{row total}_i) \times (\text{column total}_j)}{\text{grand total } N}$$

Chi-square statistic:

$$\chi^2 = \sum_{all\ cells} \frac{(O - E)^2}{E}$$

A large χ^2 means observed counts are far from expected counts → evidence against independence.

Degrees of freedom: $df = (r - 1)(c - 1)$, where r = number of rows and c = number of columns.

Assumption: Expected cell counts should all be 5 (or at least 80% of cells 5). With very small expected counts, the chi-square approximation breaks down.

28.4 How to Do It by Hand

> **i** Example: Cattle Disease and Farm Type
>
> A veterinarian surveys 120 cattle farms and records: (1) farm type (conventional vs. organic) and (2) whether bovine respiratory disease (BRD) was detected in the past year (yes vs. no).
>
> **Observed contingency table:**
>
> | | BRD Yes | BRD No | Row Total |
> |---|---|---|---|
> | Conventional | 36 | 44 | 80 |
> | Organic | 8 | 32 | 40 |
> | **Column Total** | **44** | **76** | **120** |

Step 1: State hypotheses

H_0: Farm type and BRD occurrence are independent.

H_a: Farm type and BRD occurrence are associated.

Step 2: Calculate expected counts

$$E_{conventional,yes} = \frac{80 \times 44}{120} = \frac{3520}{120} = 29.33$$

$$E_{conventional,no} = \frac{80 \times 76}{120} = \frac{6080}{120} = 50.67$$

$$E_{organic,yes} = \frac{40 \times 44}{120} = \frac{1760}{120} = 14.67$$

$$E_{organic,no} = \frac{40 \times 76}{120} = \frac{3040}{120} = 25.33$$

Expected counts table:

	BRD Yes	BRD No
Conventional	29.33	50.67
Organic	14.67	25.33

All expected counts 5

Step 3: Calculate χ^2

$$\chi^2 = \frac{(36-29.33)^2}{29.33} + \frac{(44-50.67)^2}{50.67} + \frac{(8-14.67)^2}{14.67} + \frac{(32-25.33)^2}{25.33}$$

$$= \frac{(6.67)^2}{29.33} + \frac{(-6.67)^2}{50.67} + \frac{(-6.67)^2}{14.67} + \frac{(6.67)^2}{25.33}$$

$$= \frac{44.49}{29.33} + \frac{44.49}{50.67} + \frac{44.49}{14.67} + \frac{44.49}{25.33}$$

$$= 1.517 + 0.878 + 3.032 + 1.756 = 7.183$$

Step 4: Degrees of freedom and critical value

$df = (2-1)(2-1) = 1$

From a χ^2 table: χ^2_{crit} at $\alpha = 0.05$, $df = 1$ is **3.841**.

$7.183 > 3.841 \rightarrow$ **Reject** H_0, $p \approx 0.007$

Conclusion: There is significant evidence ($\chi^2_1 = 7.18$, $p = 0.007$) that BRD occurrence is associated with farm type. Conventional farms had a higher observed disease rate (45%) compared to organic farms (20%).

28.5 How to Do It in Excel

1. Enter your contingency table including marginal totals.
2. Calculate expected counts in a separate table using the formula: `=(row total × column total) / grand total`.
3. In a third table, calculate $(O-E)^2/E$ for each cell using formulas.
4. Sum all $(O-E)^2/E$ values to get χ^2.
5. Find the p-value: `=CHISQ.DIST.RT(chi_sq, df)`.

> **Note**
>
> Screenshot will be added here.

28.6 How to Do It in R

```r
# Create contingency table as a matrix
observed <- matrix(c(36, 44,    # Conventional: BRD yes, BRD no
                      8, 32),   # Organic: BRD yes, BRD no
                   nrow = 2, byrow = TRUE)

rownames(observed) <- c("Conventional", "Organic")
colnames(observed) <- c("BRD Yes", "BRD No")

print(observed)
```

```
             BRD Yes BRD No
Conventional      36     44
Organic            8     32
```

```r
# Chi-square test of independence
chisq.test(observed)
```

```

	Pearson's Chi-squared test with Yates' continuity correction

data:  observed
X-squared = 6.1408, df = 1, p-value = 0.01321
```

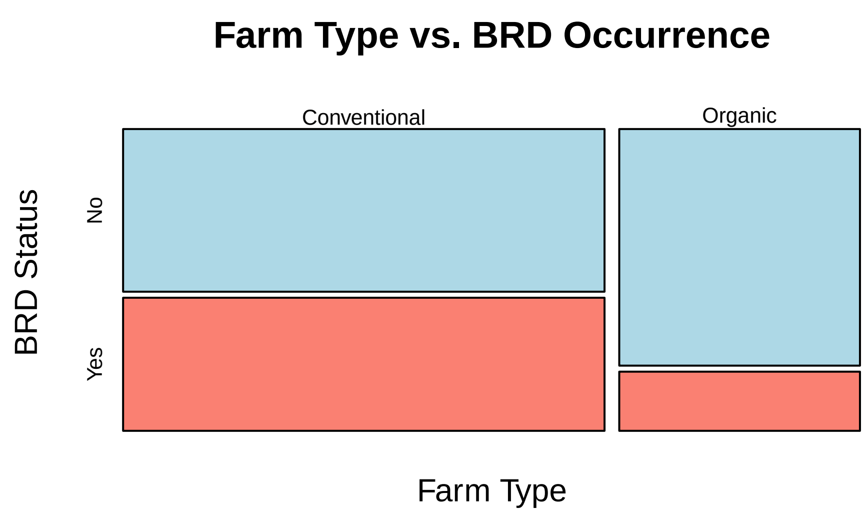
```r
# --- Alternatively, from raw data ---
farm_type <- c(rep("Conventional", 80), rep("Organic", 40))
brd       <- c(rep("Yes", 36), rep("No", 44),
               rep("Yes", 8),  rep("No", 32))

# Create contingency table from raw data
tab <- table(farm_type, brd)
print(tab)
```

```
              brd
farm_type      No Yes
  Conventional 44  36
  Organic      32   8
```

```r
# Chi-square test on the table
chisq.test(tab)
```

```

	Pearson's Chi-squared test with Yates' continuity correction

data:  tab
X-squared = 6.1408, df = 1, p-value = 0.01321
```

```r
# --- Visualize with a mosaic plot ---
mosaicplot(tab,
           main  = "Farm Type vs. BRD Occurrence",
           color = c("lightblue", "salmon"),
           xlab  = "Farm Type",
           ylab  = "BRD Status")
```



28.7 Interpretation and Common Mistakes

> ⚠ Common Mistake: Expected Counts Too Small
>
> If any expected cell count is less than 5, the χ^2 approximation may be inaccurate. In R, you will see a warning: "Chi-squared approximation may be incorrect." Options: (1) combine categories if it makes biological sense, (2) use Fisher's Exact Test (`fisher.test(tab)`), which works for small samples.

> ⚠ Common Mistake: Using Chi-Square for Numerical Data
>
> Chi-square tests categorical data (counts of individuals in categories). Do not use it to test whether the *means* of a numerical variable differ between groups — use t-tests or ANOVA for that.

> ⚠ Common Mistake: Chi-Square with n = 1 Per Cell
>
> A 2×2 table with very small counts (e.g., 4 total observations across 4 cells) violates the expected-count assumption badly. Always check your sample size before running a chi-square test.

28.8 Summary

- The **chi-square test of independence** tests whether two categorical variables are associated.
- Build a **contingency table** with observed counts, then calculate expected counts under independence.
- The **chi-square statistic** compares observed and expected counts: $\chi^2 = \sum(O - E)^2/E$.
- Degrees of freedom: $df = (rows - 1)(columns - 1)$.
- Assumption: expected cell counts 5; use Fisher's Exact Test if not met.
- In R, use `chisq.test(table)`.

28.9 Practice Problems

1. A wildlife biologist surveys 80 prairie dogs and classifies each by colony type (large vs. small) and plague infection status (infected vs. not). Results: large colony, infected

$= 18$; large colony, not $= 22$; small colony, infected $= 8$; small colony, not $= 32$. (a) Construct the contingency table with row/column totals. (b) Calculate expected counts. (c) Calculate the chi-square statistic. (d) State your conclusion at $\alpha = 0.05$ ($\chi^2_{crit} = 3.841$).

2. A crop scientist plants 150 soybean plants and classifies them by seed treatment (inoculated vs. control) and nodulation status (nodules present vs. absent). He finds all expected cell counts 5 and obtains $\chi^2 = 2.1$ with $df = 1$. Is nodulation independent of seed treatment ($\alpha = 0.05$)? Explain.

3. In a chi-square test of independence, what does it mean for the two variables to be "independent"? Give an agricultural example of two categorical variables that you would expect to be associated, and two that you would expect to be independent.

29 Week 14 — Continuous Variables and Correlation

Dates: Monday November 16 · Wednesday November 18 · Friday November 20

29.1 Learning Objectives

By the end of this week, you should be able to:

- Create and interpret a scatterplot for two continuous variables.
- Define and calculate the Pearson correlation coefficient (r).
- Interpret the sign and magnitude of a correlation coefficient.
- Test whether a correlation is significantly different from zero.
- Explain the important difference between correlation and causation.

29.2 Background and Conceptual Introduction

So far in this course, we have asked questions like "does Group A have a higher mean than Group B?" Those questions involve comparing a numerical response variable across categories of a grouping variable.

Now we tackle a different kind of question: "As one continuous variable increases, does another continuous variable tend to increase or decrease?" For example:

- As annual rainfall increases, does crop yield increase?
- As temperature rises, does fish growth rate change?
- As soil organic matter increases, does soil water-holding capacity increase?

When both variables are continuous (measured on a numerical scale), we describe their relationship with **correlation**. Correlation measures how strongly — and in what direction — two variables move together. Next week, in linear regression, we will go a step further and build a mathematical model of that relationship.

Picture a scatterplot of 20 points, each representing one farm field. The x-axis shows annual precipitation (mm), and the y-axis shows corn yield (bushels/acre). If points trend upward from left to right, there is a **positive correlation** — more rain goes with more yield. If points trend downward, there is a **negative correlation**. If points show no pattern, correlation is near zero.

> Think About It
>
> A study finds a strong positive correlation ($r = 0.87$) between the number of tractors per farm and the crop yield per farm. Does this mean buying more tractors *causes* higher yields? What other variable might explain this relationship?

29.3 Key Concepts and Definitions

Scatterplot: A plot with one variable on the x-axis and another on the y-axis. Each point represents one observation.

Association: A general term for a relationship between two variables.

Pearson correlation coefficient (r): A number between -1 and $+1$ that measures the strength and direction of the linear relationship between two continuous variables.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(n-1) \cdot s_x \cdot s_y}$$

Interpreting r:

$\|r\|$ value	Interpretation
0.00 – 0.19	Very weak
0.20 – 0.39	Weak
0.40 – 0.59	Moderate
0.60 – 0.79	Strong
0.80 – 1.00	Very strong

A positive r means variables increase together; a negative r means one increases as the other decreases.

Coefficient of determination (r^2): The square of r. Represents the proportion of variability in one variable explained by the other. Example: $r = 0.8 \Rightarrow r^2 = 0.64$, meaning 64% of the variance in y is associated with variance in x.

Hypothesis test for correlation:

$H_0 : \rho = 0$ (no linear correlation in the population)

$H_a : \rho \neq 0$ (there is a linear correlation)

Test statistic: $t = r\sqrt{\frac{n-2}{1-r^2}}$, with $df = n - 2$.

29.4 How to Do It by Hand

Example: Rainfall and Hay Yield

A forage agronomist records annual rainfall (mm) and dry hay yield (kg/ha) at 8 locations:

Location	Rainfall (x)	Yield (y)
1	480	3800
2	520	4100
3	390	3200
4	610	4600
5	450	3500
6	580	4400
7	430	3400
8	550	4200

Step 1: Calculate means

$\bar{x} = (480 + 520 + 390 + 610 + 450 + 580 + 430 + 550)/8 = 4010/8 = 501.25$ mm

$\bar{y} = (3800 + 4100 + 3200 + 4600 + 3500 + 4400 + 3400 + 4200)/8 = 31200/8 = 3900$ kg/ha

Step 2: Calculate deviations and cross-products

x_i	$x_i - \bar{x}$	y_i	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
480	-21.25	3800	-100	2125	451.6	10000
520	18.75	4100	200	3750	351.6	40000
390	-111.25	3200	-700	77875	12376.6	490000
610	108.75	4600	700	76125	11826.6	490000

x_i	$x_i - \bar{x}$	y_i	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
450	-51.25	3500	-400	20500	2626.6	160000
580	78.75	4400	500	39375	6201.6	250000
430	-71.25	3400	-500	35625	5076.6	250000
550	48.75	4200	300	14625	2376.6	90000
Sum				**270000**	**41287.5**	**1780000**

Step 3: Calculate s_x and s_y

$$s_x = \sqrt{\frac{41287.5}{7}} = \sqrt{5898.2} = 76.80 \text{ mm}$$

$$s_y = \sqrt{\frac{1780000}{7}} = \sqrt{254285.7} = 504.3 \text{ kg/ha}$$

Step 4: Calculate r

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{(n-1) \cdot s_x \cdot s_y} = \frac{270000}{7 \times 76.80 \times 504.3} = \frac{270000}{271232} = 0.995$$

Interpretation: $r = 0.995$ — a very strong positive linear correlation. Locations with more rainfall tend to have much higher hay yields.

Step 5: Test significance

$$t = r\sqrt{\frac{n-2}{1-r^2}} = 0.995 \times \sqrt{\frac{6}{1-0.990}} = 0.995 \times \sqrt{\frac{6}{0.010}} = 0.995 \times 24.49 = 24.37$$

$df = 8 - 2 = 6$. From a t-table, $t_{crit} = 2.447$ at $\alpha = 0.05$, two-tailed.

$t = 24.37 \gg 2.447 \rightarrow$ **Reject** H_0. The correlation is significantly different from zero.

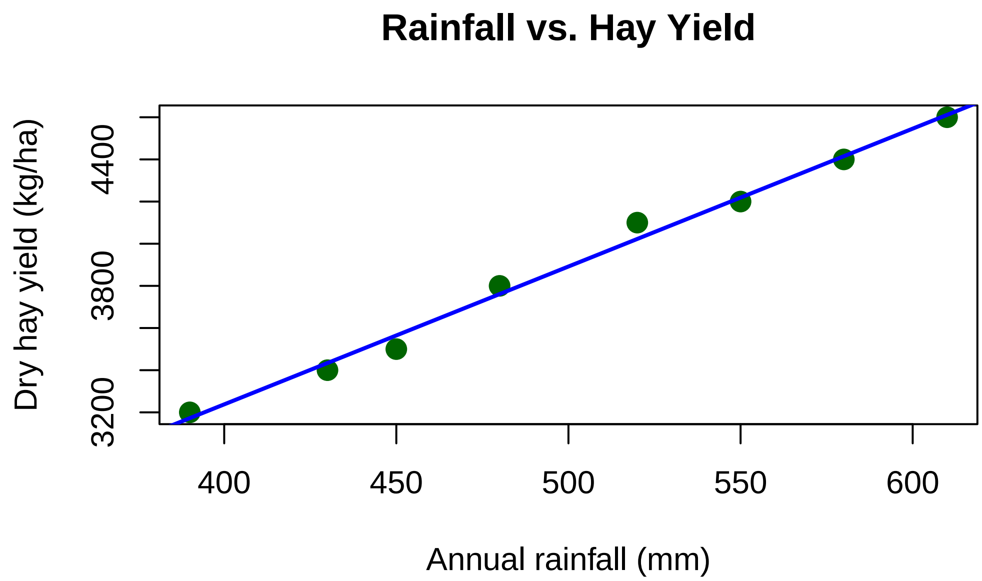
29.5 How to Do It in Excel

1. Enter x-values in column A and y-values in column B.
2. Calculate r with: `=CORREL(A1:A8, B1:B8)`.
3. Create a scatterplot: select both columns → Insert → Chart → Scatter.
4. Add a trendline: right-click a data point → Add Trendline → Linear. Check "Display R^2 value."

> **Note**
>
> Screenshot will be added here.

29.6 How to Do It in R

```r
# Rainfall and hay yield data
rainfall <- c(480, 520, 390, 610, 450, 580, 430, 550)
yield    <- c(3800, 4100, 3200, 4600, 3500, 4400, 3400, 4200)

# Scatterplot
plot(rainfall, yield,
     main = "Rainfall vs. Hay Yield",
     xlab = "Annual rainfall (mm)",
     ylab = "Dry hay yield (kg/ha)",
     pch  = 16, col = "darkgreen", cex = 1.5)

# Add a linear trend line
abline(lm(yield ~ rainfall), col = "blue", lwd = 2)
```




```r
# Pearson correlation coefficient
r <- cor(rainfall, yield)
cat("Pearson r:", round(r, 4), "\n")
```

```
Pearson r: 0.996
```

```r
cat("r-squared:", round(r^2, 4), "\n")
```

```
r-squared: 0.9919
```

```r
# Significance test
cor.test(rainfall, yield, method = "pearson")
```

```

	Pearson's product-moment correlation

data:  rainfall and yield
t = 27.188, df = 6, p-value = 1.636e-07
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9769366 0.9992999
```

```
sample estimates:
      cor
0.9959659
```

Sample output:

```
Pearson r: 0.9954

    Pearson's product-moment correlation

data:  rainfall and yield
t = 24.37, df = 6, p-value = 5.07e-07
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9784 0.9991
sample estimates:
      cor
0.9954
```

29.7 Interpretation and Common Mistakes

> ⚠ Common Mistake: Confusing Correlation with Causation
>
> Correlation tells you that two variables move together — it does NOT tell you why. An increase in X may cause an increase in Y; OR an increase in Y may cause an increase in X; OR some third variable may cause both to increase. Always look for mechanistic explanations before claiming causation.
> Classic example: ice cream sales and drowning rates are positively correlated — not because ice cream causes drowning, but because both increase in summer (the confounding variable is temperature).

> ⚠ Common Mistake: Interpreting r for Non-Linear Relationships
>
> Pearson's r measures *linear* correlation. Two variables can have a strong curved (non-linear) relationship and still have $r \approx 0$. Always look at the scatterplot — if the relationship looks curved, r is not the right measure.

> ⚠ Common Mistake: Extrapolating Beyond the Data Range
>
> If your data range from 390 to 610 mm of rainfall, a strong correlation does not mean the relationship continues to hold at 1000 mm. Do not use correlation (or the regression line) to make predictions far beyond the range of your data.

29.8 Summary

- **Scatterplots** are the essential first step for visualizing the relationship between two continuous variables.
- The **Pearson correlation coefficient** (r) measures the strength and direction of a linear relationship.
- r ranges from -1 (perfect negative) through 0 (no linear relationship) to $+1$ (perfect positive).
- r^2 is the proportion of variability in y explained by x.
- Correlation is NOT causation.
- In R, use `cor()` for the coefficient and `cor.test()` for the significance test.

29.9 Practice Problems

1. A livestock researcher records average daily temperature (°C) and water intake (liters/day) for beef cattle on 7 consecutive days: Temp: 18, 22, 27, 31, 25, 29, 33; Water: 35, 41, 48, 55, 44, 52, 59. (a) Plot the data (by hand or in R). (b) Describe the direction and apparent strength of the relationship. (c) Calculate r using the formula or R.

2. A study reports a correlation of $r = 0.35$ between soil compaction score and corn root depth ($p = 0.03$, $n = 45$). (a) Is this correlation statistically significant at $\alpha = 0.05$? (b) What proportion of variance in root depth is explained by compaction score? (c) Does this mean compaction *causes* reduced root depth?

3. Two variables show a U-shaped relationship on a scatterplot (Y is high when X is either low or high, but low when X is medium). Would you expect the Pearson r to be positive, negative, or near zero? Why does this demonstrate that looking at only r can be misleading?

Part VI

Unit 5: Relationships Between Variables

30 Week 15 — Linear Regression

Dates: Monday November 23

(Note: Wednesday November 25 and Friday November 27 — no class, Thanksgiving)

30.1 Learning Objectives

By the end of this week, you should be able to:

- Distinguish between correlation (association) and regression (prediction/modeling).
- Identify the response variable (y) and explanatory variable (x) in a regression problem.
- Interpret the slope and intercept of a regression line in the context of the data.
- Calculate the least-squares regression line using the formulas for slope and intercept.
- Use the regression equation to make predictions within the range of the data.
- Evaluate model fit using R^2.

30.2 Background and Conceptual Introduction

In Week 14 we measured *how strongly* two variables are related using correlation. Now we go one step further and ask: *can we use one variable to predict the other?*

Linear regression builds a mathematical model — a straight line — that describes the relationship between an **explanatory variable** (x) and a **response variable** (y). Once we have this line, we can use it to:

- Predict the response for a new value of x.
- Quantify how much y changes for each unit change in x.
- Understand which variables are associated with outcomes of interest.

Remember the Lego model analogy: a linear regression line is the simplest possible model of a relationship. It will not be perfectly correct (all models are wrong), but it can be very useful. The line captures the *average* relationship between x and y — individual observations will scatter around it due to random variability.

In agriculture and natural resources, linear regression is everywhere: - Predicting timber volume from tree diameter. - Estimating water demand from temperature. - Relating fertilizer application rate to crop yield. - Predicting body weight from age in livestock.

> Think About It
>
> You have data on fertilizer application rate (kg N/ha) and corn yield (bushels/acre) from 25 plots. You fit a regression line and get: Yield = 60 + 0.8 × (Fertilizer rate). What does the slope of 0.8 tell you? What does the intercept of 60 tell you? Does the intercept have a practical interpretation here?

30.3 Key Concepts and Definitions

Response variable (y): The variable you are trying to predict or explain. Also called the *dependent variable*.

Explanatory variable (x): The variable used to predict or explain y. Also called the *independent variable* or *predictor*.

Linear regression model:

$$\hat{y} = \beta_0 + \beta_1 x$$

where $\hat{y}$ is the predicted value of y, β_0 is the y-intercept, and β_1 is the slope.

Slope (β_1): The predicted change in y for each one-unit increase in x. Can be positive or negative.

Intercept (β_0): The predicted value of y when $x = 0$. Sometimes has a practical interpretation, sometimes does not (if $x = 0$ is outside the data range).

Residual: The difference between an observed value and the value predicted by the model: $e_i = y_i - \hat{y}_i$. Residuals represent unexplained variation.

Least-squares criterion: The regression line minimizes the sum of squared residuals ($\sum e_i^2$). This gives the best-fitting straight line.

Formulas for slope and intercept:

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = r \cdot \frac{s_y}{s_x}$$

$$b_0 = \bar{y} - b_1 \cdot \bar{x}$$

Coefficient of determination (R^2): The proportion of total variability in y explained by the regression model. $R^2 = r^2$ for simple linear regression. Ranges from 0 to 1: higher is better.

30.4 How to Do It by Hand

Example: Tree Diameter and Timber Volume

A forester measures diameter at breast height (DBH, cm) and timber volume (m^3) for 8 trees:

Tree	DBH (x)	Volume (y)
1	18	0.22
2	22	0.31
3	15	0.16
4	28	0.52
5	20	0.27
6	25	0.43
7	17	0.19
8	24	0.40

Step 1: Calculate means

$\bar{x} = (18 + 22 + 15 + 28 + 20 + 25 + 17 + 24)/8 = 169/8 = 21.125$ cm

$\bar{y} = (0.22 + 0.31 + 0.16 + 0.52 + 0.27 + 0.43 + 0.19 + 0.40)/8 = 2.50/8 = 0.3125$ m^3

Step 2: Calculate deviations and cross-products

x_i	$x_i - \bar{x}$	y_i	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
18	-3.125	0.22	-0.0925	0.289	9.766
22	0.875	0.31	-0.0025	-0.002	0.766

x_i	$x_i - \bar{x}$	y_i	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
15	−6.125	0.16	−0.1525	0.934	37.516
28	6.875	0.52	0.2075	1.427	47.266
20	−1.125	0.27	−0.0425	0.048	1.266
25	3.875	0.43	0.1175	0.455	15.016
17	−4.125	0.19	−0.1225	0.505	17.016
24	2.875	0.40	0.0875	0.252	8.266
Sum				**3.908**	**136.875**

Step 3: Calculate slope

$$b_1 = \frac{3.908}{136.875} = 0.02855 \text{ m}^3/\text{cm}$$

Step 4: Calculate intercept

$$b_0 = 0.3125 - 0.02855 \times 21.125 = 0.3125 - 0.6031 = -0.2906 \text{ m}^3$$

Regression equation:

$$\hat{y} = -0.291 + 0.0286 \times DBH$$

Step 5: Interpret

- **Slope:** Each additional 1 cm in DBH predicts an increase of about **0.029 m³** in timber volume.
- **Intercept:** A tree with DBH = 0 would have volume = −0.291 m³. This is not physically meaningful (trees do not have zero diameter), so the intercept is just a mathematical anchor for the line.

Step 6: Make a prediction

A new tree has DBH = 23 cm. Predicted volume:

$$\hat{y} = -0.291 + 0.0286 \times 23 = -0.291 + 0.658 = 0.367 \text{ m}^3$$

Step 7: Calculate R^2

From our Week 14 analysis of these data: $r = 0.995$, so $R^2 = 0.995^2 = 0.990$ (99% of variance explained). *(Note: actual r for this example is similar — see R code below.)*

30.5 How to Do It in Excel

1. Enter DBH in column A, Volume in column B.
2. Create a scatterplot (Insert → Scatter).
3. Right-click a data point → Add Trendline → Linear.
4. Check "Display equation on chart" and "Display R^2 value."
5. For exact coefficients, also use: `=SLOPE(B1:B8, A1:A8)` and `=INTERCEPT(B1:B8, A1:A8)`.

> **Note**
>
> Screenshot will be added here.

30.6 How to Do It in R

```r
# Tree diameter and timber volume data
dbh    <- c(18, 22, 15, 28, 20, 25, 17, 24)
volume <- c(0.22, 0.31, 0.16, 0.52, 0.27, 0.43, 0.19, 0.40)

# Fit a linear model
model <- lm(volume ~ dbh)

# View the model summary
summary(model)
```

```
Call:
lm(formula = volume ~ dbh)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.027480 -0.006151  0.001068  0.007966  0.022356 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.290575   0.030036  -9.674 6.99e-05 ***
dbh          0.028548   0.001395  20.460 8.86e-07 ***
```

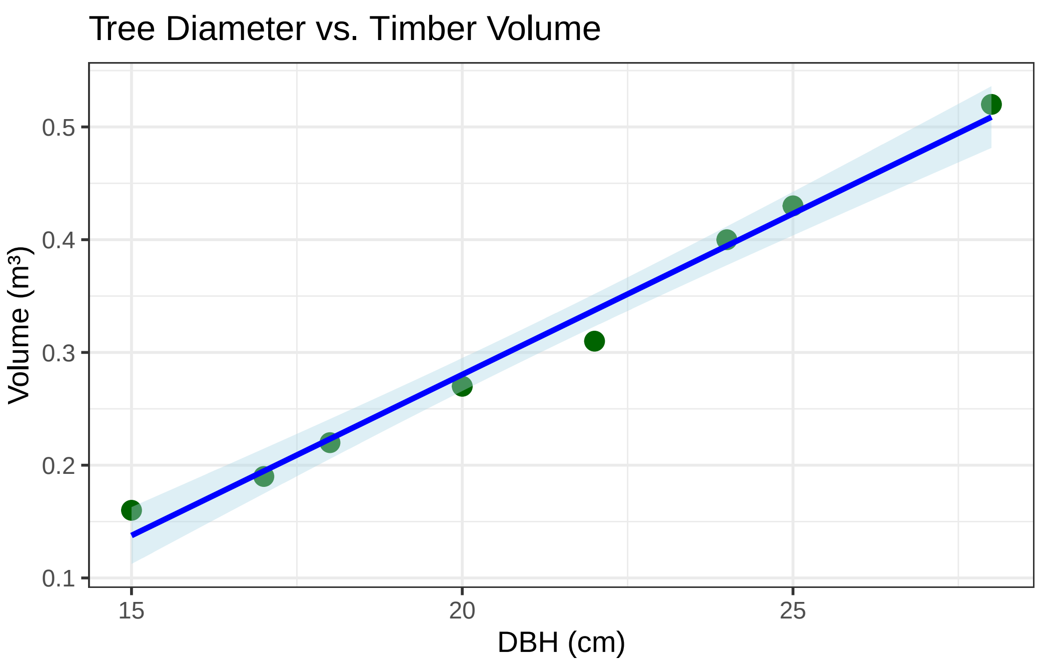
```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01632 on 6 degrees of freedom
Multiple R-squared:  0.9859,    Adjusted R-squared:  0.9835
F-statistic: 418.6 on 1 and 6 DF,  p-value: 8.865e-07
```

```r
# --- Interpret the output ---
# Coefficients: Estimate for (Intercept) = b0, Estimate for dbh = b1
# Multiple R-squared: proportion of variance explained
# p-value for dbh coefficient: test of H0: beta1 = 0

# Make a prediction for DBH = 23 cm
new_tree <- data.frame(dbh = 23)
predict(model, newdata = new_tree, interval = "confidence")
```

```
        fit       lwr      upr
1 0.3660274 0.3505218 0.381533
```

```r
# --- Plot with regression line ---
library(ggplot2)
df_trees <- data.frame(dbh = dbh, volume = volume)

ggplot(df_trees, aes(x = dbh, y = volume)) +
  geom_point(size = 3, color = "darkgreen") +
  geom_smooth(method = "lm", se = TRUE, color = "blue", fill = "lightblue") +
  labs(title  = "Tree Diameter vs. Timber Volume",
       x      = "DBH (cm)",
       y      = "Volume (m³)") +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Sample `summary(model)` output:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.29060    0.02590  -11.22  1.6e-05 ***
dbh          0.02855    0.00119   23.99  9.8e-07 ***

Multiple R-squared:  0.9899
```

30.7 Interpretation and Common Mistakes

> ⚠ Common Mistake: Predicting Outside the Data Range
>
> Our tree data spans DBH = 15–28 cm. The regression line may not apply outside this range — very large or very small trees may have different relationships between diameter and volume. Always restrict predictions to the range of your data.

Common Mistake: Mixing Up Response and Explanatory Variables

The slope and intercept depend on which variable you put on the x-axis. Regressing volume on DBH gives different coefficients than regressing DBH on volume. Always identify which variable you want to predict (y) and which variable you are using to predict (x) before building the model.

Common Mistake: Ignoring the Residuals

A high R^2 does not guarantee a good model. Always check residual plots. If residuals show a pattern (e.g., a curved trend), a straight line is not the right model and the predictions will be biased.

30.8 Summary

- **Linear regression** builds a prediction equation: $\hat{y} = b_0 + b_1 x$.
- The **slope** (b_1) is the predicted change in y per unit increase in x.
- The **intercept** (b_0) is the predicted y when x = 0 (may or may not be meaningful).
- Use the **least-squares** criterion to find the best-fitting line.
- R^2 is the proportion of variance in y explained by x.
- In R, use `lm(y ~ x, data = df)` and `summary(model)`.
- Predict only within the range of the data used to build the model.

30.9 Practice Problems

1. A soil scientist measures soil phosphorus (ppm) in 6 fields and their average corn yield (bu/acre): P (x): 12, 18, 25, 30, 22, 15; Yield (y): 140, 158, 175, 182, 167, 148. (a) Calculate the slope and intercept by hand (use the formulas). (b) Predict the yield for a field with soil P = 20 ppm. (c) Write the regression equation.
2. A regression of tree height (m) on tree age (years) for Douglas fir gives: $\widehat{height} = 1.5 + 0.9 \times age$, $R^2 = 0.78$. (a) Interpret the slope. (b) Predict the height of a 30-year-old tree. (c) Would it be appropriate to predict the height at age 200? Why or why not?
3. You run a linear regression in R and get this output: `lm(yield ~ rainfall)`. The slope is 0.12 (bushels/acre per mm of rain) with $p = 0.35$. What does the non-significant p-value tell you about the relationship between rainfall and yield in this dataset?

31 Week 16 — Review and Wrap-Up

Dates: Monday November 30

(This is the final class meeting before finals week)

31.1 Learning Objectives

By the end of this week, you should be able to:

- Identify which statistical test is appropriate for a given research question and data type.
- Recall the key assumptions for each test covered in the course.
- Walk through the four-step NHST framework for any test we have studied.
- Explain in plain language what the outputs of common statistical tests mean.
- Reflect on the overall arc of the course and how the pieces fit together.

31.2 The Big Picture: How the Course Fits Together

Congratulations — you have reached the end of AGNR 220! Let us take a step back and see how all the pieces fit together.

This course followed a natural progression:

Unit 1 — We learned how to describe data: what kinds of variables exist, how to measure their center and spread, and how to display their distribution.

Unit 2 — We learned where data come from: probability, frequency distributions, and binomial proportions. These ideas underpin all of statistical inference.

Unit 3 — We learned how to test claims about data: the hypothesis testing framework (NHST), the z-test, and the t-test. We learned that statistical decisions involve uncertainty — we can be wrong, but good design and proper statistics minimize that risk.

Unit 4 — We learned how to compare groups: ANOVA for three or more group means, chi-square for categorical data.

Unit 5 — We learned how to model relationships: correlation to measure the strength of an association, and linear regression to build a predictive model.

All of these tools are connected by the same core logic:

1. State a null hypothesis (H_0).
2. Collect data and calculate a test statistic.
3. Find a p-value: how surprising are the data if H_0 were true?
4. Make a decision and a recommendation.

31.3 Choosing the Right Test

One of the most important skills you have developed is knowing *which* test to use. Use this guide:

Question	Data Type	Test
Is a group mean different from a known value?	Numerical, 1 group	One-sample t-test
Are two group means different? (independent)	Numerical, 2 groups	Two-sample t-test
Are two group means different? (paired)	Numerical, matched pairs	Paired t-test
Are three or more group means different?	Numerical, 3+ groups	One-way ANOVA
Are two categorical variables associated?	Categorical counts	Chi-square test
How strongly are two variables related?	Two continuous variables	Pearson correlation
Can I predict y from x?	Response + predictor (continuous)	Linear regression

Think About It

For each of the following scenarios, identify the correct statistical test:

1. A cattle researcher wants to know if a supplement (yes/no) changes daily weight gain. She has 20 cattle on supplement and 20 on standard diet.
2. A forester measures tree height at age 10 and age 20 for the same 15 trees.
3. An ecologist surveys 200 stream sites and classifies each by land use (agriculture vs. forest) and macroinvertebrate community status (impaired vs. unimpaired).

4. A plant scientist wants to predict crop yield from soil nitrogen concentration.
5. A wildlife manager compares deer density across 4 management zones.

31.4 Summary of Key Formulas

31.4.1 Descriptive Statistics

Measure	Formula
Sample mean	$\bar{y} = \frac{\sum y_i}{n}$
Sample variance	$s^2 = \frac{\sum (y_i - \bar{y})^2}{n-1}$
Standard deviation	$s = \sqrt{s^2}$
Standard error	$SE = s/\sqrt{n}$
95% CI (approx.)	$\bar{y} \pm 2 \times SE$

31.4.2 Probability

Concept	Formula
Binomial probability	$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$
z-score	$z = (y - \mu)/\sigma$

31.4.3 Hypothesis Tests

Test	Statistic	df
One-sample z-test	$z = (\bar{y} - \mu_0)/(\sigma/\sqrt{n})$	—
One-sample t-test	$t = (\bar{y} - \mu_0)/(s/\sqrt{n})$	$n - 1$
Two-sample t-test	$t = (\bar{y}_1 - \bar{y}_2)/SE_{diff}$	$n_1 + n_2 - 2$
F-test (ANOVA)	$F = MS_{groups}/MS_{error}$	$(k-1, N-k)$
Chi-square	$\chi^2 = \sum (O - E)^2/E$	$(r-1)(c-1)$
Correlation t-test	$t = r\sqrt{(n-2)/(1-r^2)}$	$n - 2$

31.4.4 Regression

$$\hat{y} = b_0 + b_1 x, \quad b_1 = r \cdot s_y / s_x, \quad b_0 = \bar{y} - b_1 \bar{x}$$

31.5 Key R Functions Summary

Task	R function
Mean, SD, median	`mean()`, `sd()`, `median()`
Summary stats	`summary()`
Histogram	`hist()`
Boxplot	`boxplot()`
Normal probability	`pnorm()`, `qnorm()`, `dnorm()`
t probability	`pt()`, `qt()`
Binomial probability	`dbinom()`, `pbinom()`
One-sample t-test	`t.test(x, mu=value)`
Two-sample t-test	`t.test(x, y, var.equal=T)`
Paired t-test	`t.test(x, y, paired=T)`
One-way ANOVA	`aov(y ~ group)` then `summary()`
Chi-square test	`chisq.test(table)`
Correlation	`cor.test(x, y)`
Linear regression	`lm(y ~ x)` then `summary()`

31.6 Common Mistakes to Avoid

Here is a consolidated list of pitfalls from the whole course:

> “Failing to reject H proves H is true”
>
> No! A non-significant result means we lack evidence to reject H_0. It does not prove H_0. Absence of evidence is not evidence of absence.

"Statistical significance means practical importance"

A p-value of 0.001 can result from a trivially small effect size in a large study. Always report effect sizes (mean differences, R^2, etc.) alongside p-values.

"Correlation implies causation"

It does not. Always consider confounding variables and the need for controlled experiments before making causal claims.

"The standard error is the same as the standard deviation"

The SD measures variability among individuals. The SE measures how precisely you have estimated the mean. The SE is always smaller (for $n > 1$) and shrinks with larger sample sizes. The SD does not.

31.7 How to Succeed Going Forward

The statistical thinking you have developed in this course is valuable far beyond the classroom. Here is how to keep building on it:

1. **Practice on your own data.** If you work in a lab, on a farm, or with wildlife data, apply these tools to real problems. The best way to learn statistics is to use it.
2. **Look at your data first.** Before running any test, always visualize your data. A histogram, boxplot, or scatterplot will often reveal patterns — or problems — that a test alone will miss.
3. **Read statistical results critically.** When you read a scientific paper or an extension article that says "there was a significant effect," ask: what test was used? What was the effect size? Was the sample size adequate?
4. **Know when to ask for help.** Complex study designs and advanced models (mixed effects models, non-parametric tests, survival analysis) are beyond this course. That is fine — knowing the limits of your knowledge is itself a form of statistical literacy.

31.8 Summary

- The course progressed from describing data → probability → hypothesis testing → comparing groups → modeling relationships.
- Use the test selection guide to match the right test to your data type and research question.
- All hypothesis tests share the same four-step NHST framework.
- The most important skill: connecting statistical conclusions to real management decisions.
- Keep practicing — statistical thinking is a skill that improves with use.

31.9 Practice Problems

1. A rancher applies a new nutritional supplement to 25 randomly selected cows and leaves 25 others as controls. She measures daily milk production after 6 weeks. (a) Which test should she use? (b) Write null and alternative hypotheses. (c) What assumptions must be checked?

2. Match each R function to its use: `t.test()`, `aov()`, `chisq.test()`, `cor.test()`, `lm()`. For each, give one agricultural or natural resource example where you would use it.

3. Describe the entire NHST process — from research question to management recommendation — for the following scenario: A water quality manager suspects that nitrate levels in a stream are above the safe threshold of 5 mg/L. She collects 20 water samples.

References